POLICY GRADIENT · SOCCER GAME · DETAILED REPORT

Policy gradient on the soccer game

This study evaluates first-goal and fixed-horizon goal-reset soccer as complementary formats; neither is designated the primary result. First-goal play is tested on deterministic and random move-order boards, while goal-reset play is tested on the deterministic board. REINFORCE, A2C and PPO use standard training and fictitious play. Results are kept separate because a win means a different thing in each format. This is the detailed report; a short version is `policy_gradient.pdf`.

Summary

- On the random board the exact solver wins 67% when it starts with the ball, 39% when it starts without the ball and 53% averaged over both starts; the six learners win 5% to 14%, 0% to 1% and 2% to 8%. On the deterministic board the exact solver ties itself, so it wins 0%.
- Against a random player the learners win 61% to 96% of games (deterministic board).
- Starting from the 94 states where the exact answer has to mix, they win 11% to 15% against the exact Nash policy; the exact solver wins 50%.
- No setting tried replicates the exact solver. The best, on 10 seeds, wins 30% when it starts with the ball (A2C, fictitious play, 10,000 iterations).

3 seeds per learner unless stated, 1,000 games per opponent.

Contents

1. Win rates against the best response
2. Setup
3. Results by game condition
4. Mixed states of the random board
5. Action probabilities
6. Why fictitious play
7. Does training converge?
8. More training
9. Fictitious play with long best responses
10. Network sharing and trimming
11. Averaging pure policies
12. A larger entropy bonus
13. Training choices
14. More seeds
15. Are the two players mirror images?
16. Other kinds of games
17. Without sampling noise: exact gradient play
18. Related work
19. Reproducing the numbers

What this report adds

- Both first-goal and fixed-horizon goal-reset scoring are included as complementary conditions; neither is designated primary. The win-rate table reports both starting positions and their 50/50 average.
- The goal-reset condition runs for a fixed 100 steps, restarting after each score; it also includes A2C using the exact solver's values as a frozen critic.
- Comparisons of one shared network and of leaving out the last steps of each episode, on all three boards.
- Fictitious play over pure (argmax) snapshots, next to the usual softmax average.
- An entropy-bonus ablation (0.01 against 0.2) for every learner, training scheme and board.
- Training curves: win rate against the best response and the kickoff policy during training.
- A sweep of training choices: how many last steps to drop, snapshot interval, five times the iterations, mini-batches, replayed data, the exact critic on every board, weight averaging.
- Ten seeds with 95% intervals for the two best-looking settings.
- Rock-paper-scissors with neural-network players, with and without a larger entropy bonus.
- A related-work list of older and recent papers, checked against Google Scholar and each paper's own page where it would load.

1. Win rates against the best response

Two players face each other. At kickoff one of them has the ball and the other has to defend, and having the ball is an advantage. Each learner is trained for both players, so it is measured in both positions. **Starts with the ball**: our learner is the player who has the ball at kickoff. **Starts without the ball**: our learner is the other player. **Average of both**: the mean of the two, one fair number that does not depend on who starts with the ball.

learner	The deterministic board (A10)				The random move-order board				The goal-reset game (fixed horizon)			
	with ball	without ball	average of both	Nash gap	with ball	without ball	average of both	Nash gap	with ball	without ball	average of both	Nash gap
Exact solver	**0%**	**0%**	**0%**	**0.000**	**67%**	**39%**	**53%**	**0.000**	**0%**	**0%**	**0%**	**0.000**
REINFORCE, standard	0%	1%	0%	0.788	13%	0%	7%	0.639	0%	0%	0%	1.139
REINFORCE, fictitious play	2%	1%	2%	0.861	8%	0%	4%	0.745	0%	0%	0%	1.453
A2C, standard	2%	1%	2%	0.763	14%	1%	8%	0.605	0%	0%	0%	1.066
A2C, fictitious play	1%	2%	1%	0.888	12%	1%	7%	0.640	0%	0%	0%	1.469
PPO, standard	0%	0%	0%	0.949	9%	0%	5%	0.750	0%	0%	0%	1.491
PPO, fictitious play	0%	0%	0%	0.864	5%	0%	2%	0.708	0%	0%	0%	1.483
A2C, exact critic, standard	–	–	–	–	–	–	–	–	0%	0%	0%	0.867
A2C, exact critic, fictitious play	–	–	–	–	–	–	–	–	0%	0%	0%	1.435

The share of 1,000 games each learner wins when the opponent plays the exact best response to it, when it starts with the ball, when it starts without the ball, and averaged over both (mean of 3 seeds). The exact solver is the top row, as context rather than a target. The win rate is a check, not an objective: no learner is trained to raise it. A dash means the learner was not run on that board.

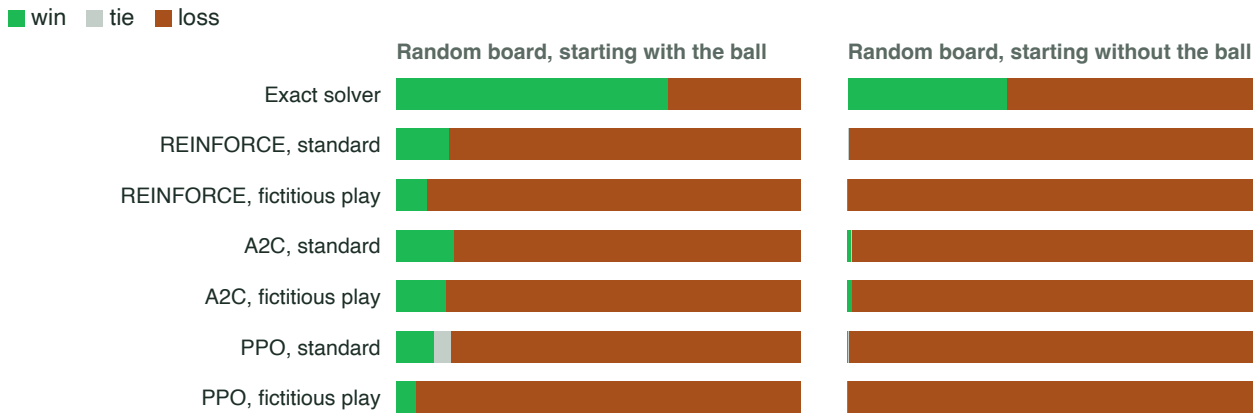
How to read the numbers

number	what it is
Win rate against the best response	The share of 1,000 games the trained player wins when the opponent plays the exact best response to it, the move that minimizes its expected discounted score difference.
Starts with the ball / starts without the ball / average of both	At kickoff one player has the ball and the other has to defend. Starts with the ball: our learner is the player who has the ball. Starts without the ball: our learner is the other player. Average of both: the mean of the two, one fair number that does not depend on who starts with the ball.
Win rate against the exact Nash policy	The same, against the exact equilibrium player.
Win rate against a random player	A sanity check: every learner should beat a player who moves at random.
A win	In first-goal play, the winner is the player who scores first. In fixed-horizon goal-reset play, the winner scores more goals over 100 steps. No goal or equal goals is a tie.
Distance from the exact mix	Total-variation distance between a learner's probabilities and the exact solver's at the first step: 0 is identical, 1 is no overlap.
Nash gap (exploitability)	The sum of both players' gains in expected discounted score if each, in turn, replaces its policy with an exact best response to the other. It is 0 at the exact Nash equilibrium; smaller is closer to equilibrium. This payoff gap evaluates the policy pair and is not a win rate.

Why the exact solver does not win 0% against the best response

The best response is the move that minimizes our player's *expected discounted score difference*, not the one that minimizes its chance of winning. At the random board's kickoff the player holding the ball is ahead: the exact solution is worth +0.15 to it even against an exact best response. Its win rate is a reference outcome, not a theoretical ceiling, because the best response minimizes discounted score rather than win probability. So the exact solver wins 67% of games and loses 33%

when it starts with the ball, and wins 39% when it starts without the ball. On the deterministic board both starts tie every game.



2. Setup

setting	value
Game formats	Both scoring formats are included as complementary conditions, with neither designated primary. First-goal termination is tested on deterministic A10 and random move-order boards. Fixed-horizon goal-reset play is tested on deterministic A10; play restarts after each score and continues for 100 steps. Results are reported separately because their outcomes differ.
Board and horizon	7×5 board, goal rows 1–3, 100 steps, discount 0.9, and remaining steps as a network input. A10 uses simultaneous moves with the carrier winning contested squares; random move-order applies moves in a random order each step.
Returns	REINFORCE: the discounted return from each step to the end of the episode, no baseline. A2C: 10-step bootstrapped advantage from a learned critic. A2C with the exact critic: the critic is the exact solver's value, frozen, so the advantage is the exact Q(s, a_0, a_1) minus V(s). PPO: GAE (λ 0.95), ratio clipped at 0.2, 4 epochs.
Updates	Saved comparison tables use one update pass on each collected REINFORCE/A2C batch. The runner now defaults to five shuffled passes (--updates-per-batch 5); use 1 to reproduce those results. PPO retains four epochs. Each pass uses all data or one shuffled mini-batch sweep. Every iteration collects 64 episodes of 100 steps; 2,000 iterations, learning rate 10^{-3}, entropy bonus 0.01, 64×64 network, softmax output, a separate network for each player. Replay remains a separate option.
Standard	Both players' current networks play each other and keep updating: real-time best responses.
Fictitious play	Each player trains against a frozen snapshot of the other, drawn at random from all its past snapshots (one every 5 iterations, a fixed budget of 5 policy-gradient iterations); the policy it reports is the average of its own snapshots. Policies, never network weights, are averaged.
Evaluation	1,000 repeated games from the kickoff against a random player, the exact Nash policy and the exact best response; wins, ties and losses counted. 3 seeds per learner unless stated.

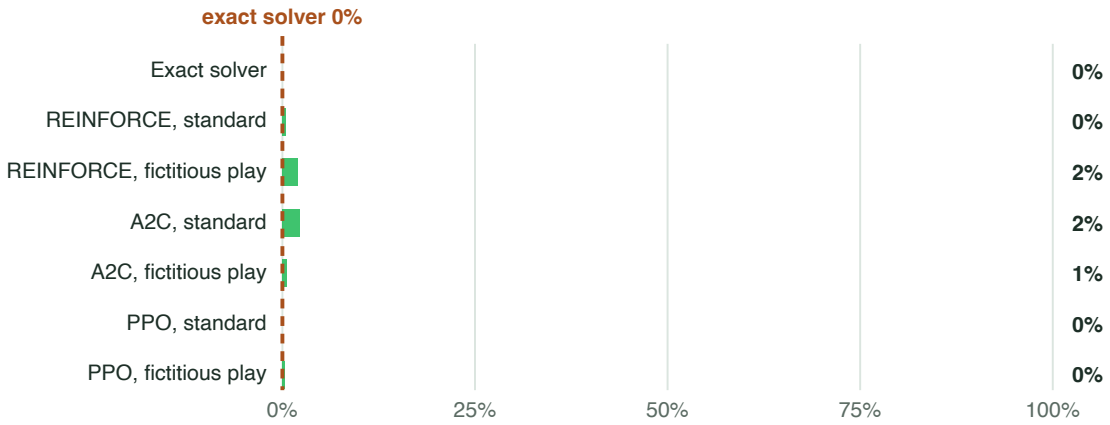
3. Results by game condition

Against the best response the exact solver wins 67% of games when it starts with the ball on the random board; the six learners win 5% to 14%. Against a random player the learners win 61% to 96% of games on the deterministic board. Fictitious-play training wins more often against a random player; neither way of training reaches the exact solver's rate against the best response.

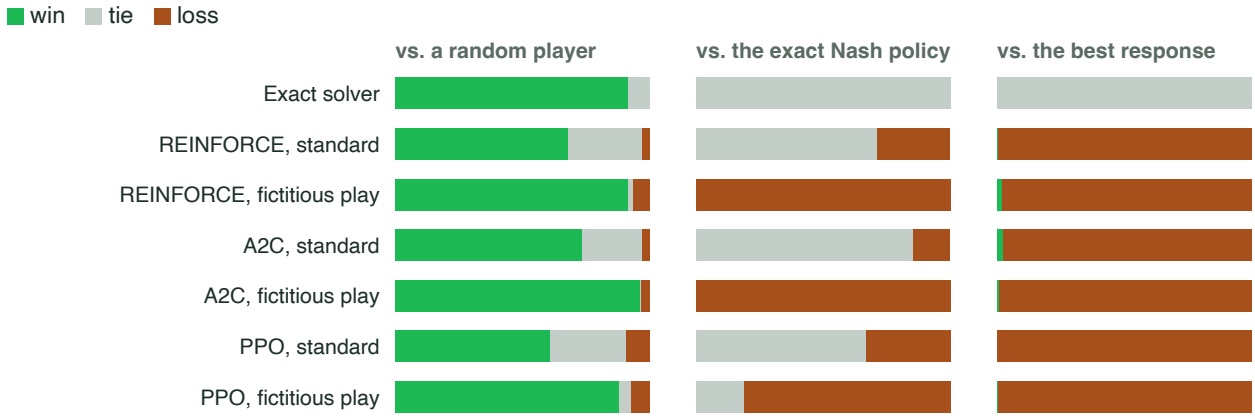
The deterministic board (A10)

Players move simultaneously and the carrier wins every contested square, so the exact solution is pure everywhere and ties itself.

Win rate against the best response (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 0%; the learners win 0% to 2%.



Wins, ties and losses against each opponent. Against the exact Nash policy the standard learners tie 67% to 85% of games and the fictitious-play learners 0% to 19%.

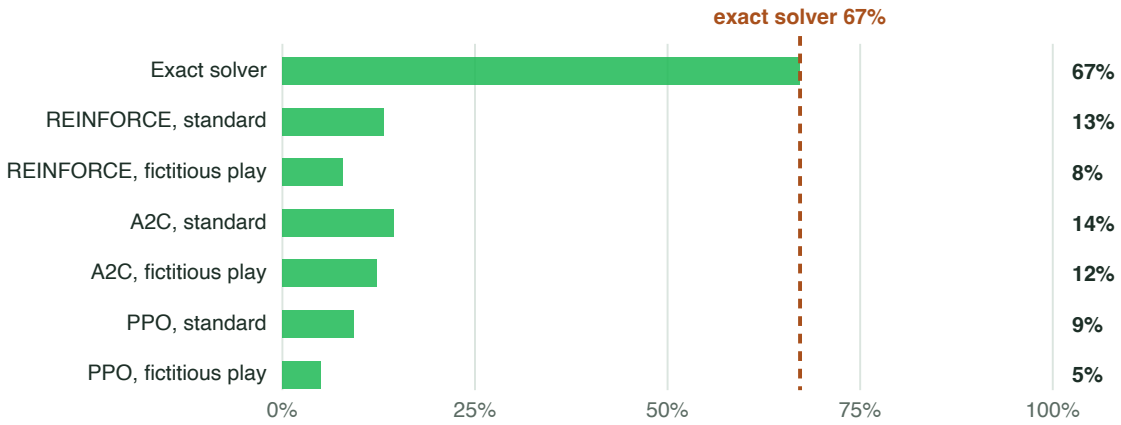


learner	vs. a random player	vs. the exact Nash policy	vs. the best response
Exact solver	**92%** tie 8% · loss 0%	**0%** tie 100% · loss 0%	**0%** tie 100% · loss 0%
REINFORCE, standard	**68%** tie 29% · loss 3%	**0%** tie 71% · loss 29%	**0%** tie 0% · loss 100%
REINFORCE, fictitious play	**92%** tie 2% · loss 7%	**0%** tie 0% · loss 100%	**2%** tie 0% · loss 98%
A2C, standard	**73%** tie 24% · loss 3%	**0%** tie 85% · loss 15%	**2%** tie 0% · loss 98%
A2C, fictitious play	**96%** tie 0% · loss 3%	**0%** tie 0% · loss 100%	**1%** tie 0% · loss 99%
PPO, standard	**61%** tie 30% · loss 9%	**0%** tie 67% · loss 33%	**0%** tie 0% · loss 100%
PPO, fictitious play	**88%** tie 5% · loss 7%	**0%** tie 19% · loss 81%	**0%** tie 0% · loss 100%

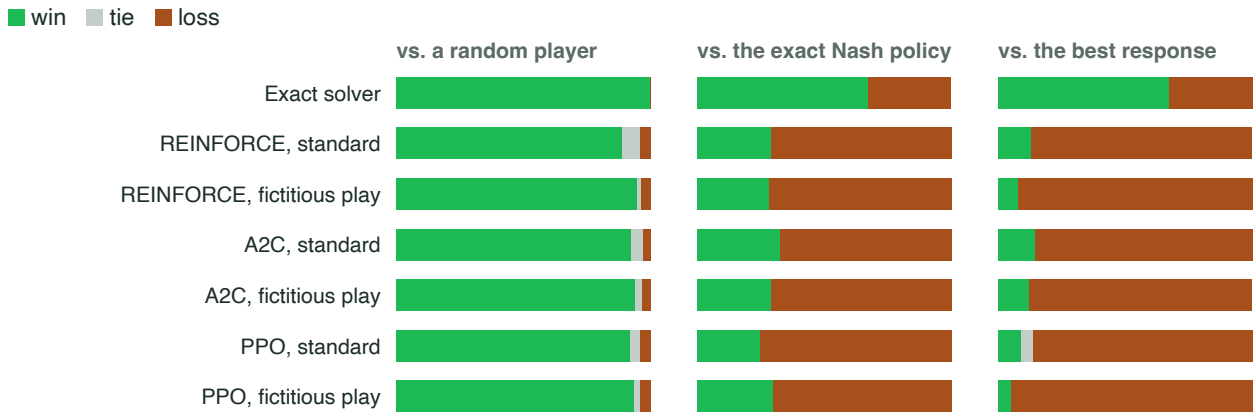
The random move-order board

Moves are applied in a random order each step; the exact solution mixes at 94 stage games.

Win rate against the best response (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 67%; the learners win 5% to 14%.



Wins, ties and losses against each opponent.

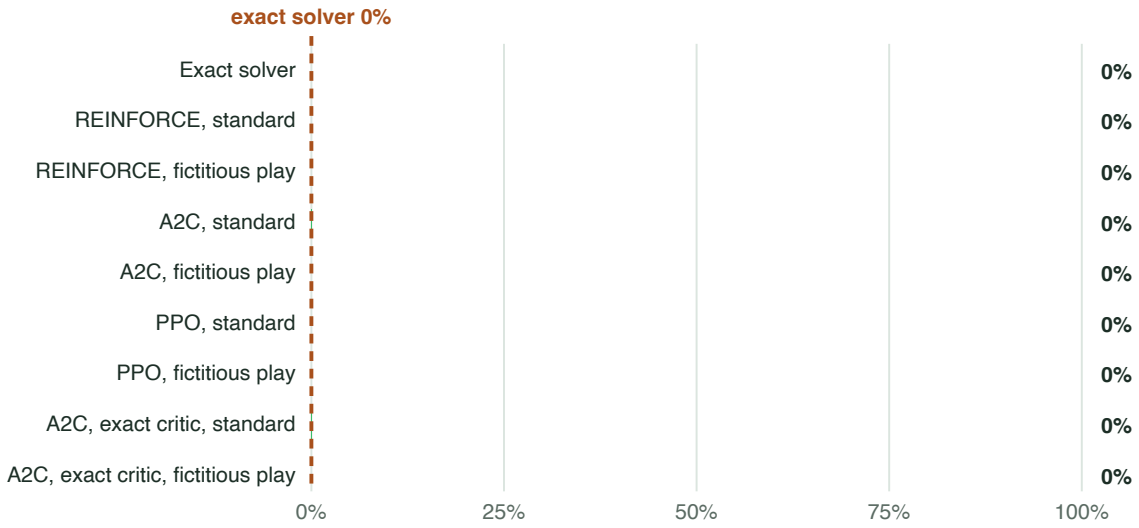


learner	vs. a random player	vs. the exact Nash policy	vs. the best response
Exact solver	**100%** tie 0% · loss 0%	**67%** tie 0% · loss 33%	**67%** tie 0% · loss 33%
REINFORCE, standard	**89%** tie 7% · loss 4%	**29%** tie 0% · loss 71%	**13%** tie 0% · loss 87%
REINFORCE, fictitious play	**95%** tie 2% · loss 4%	**28%** tie 0% · loss 72%	**8%** tie 0% · loss 92%
A2C, standard	**92%** tie 5% · loss 3%	**33%** tie 0% · loss 67%	**14%** tie 0% · loss 86%
A2C, fictitious play	**94%** tie 3% · loss 3%	**29%** tie 0% · loss 71%	**12%** tie 0% · loss 88%
PPO, standard	**92%** tie 4% · loss 4%	**25%** tie 0% · loss 75%	**9%** tie 4% · loss 86%
PPO, fictitious play	**94%** tie 2% · loss 4%	**30%** tie 0% · loss 70%	**5%** tie 0% · loss 95%

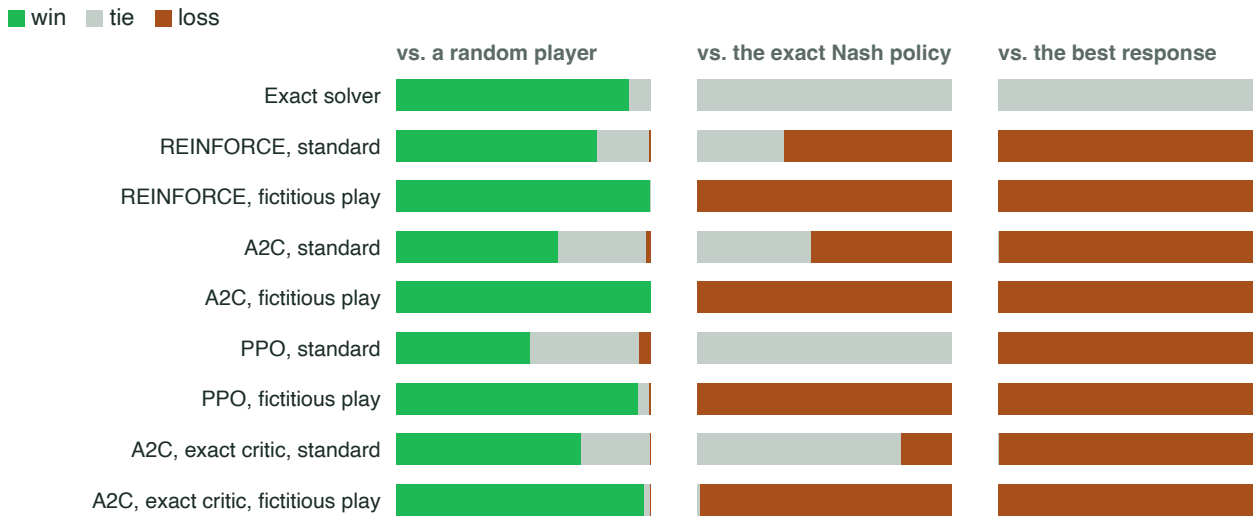
The goal-reset game (fixed horizon)

Play restarts after every goal and runs for exactly 100 steps; a win is more goals than the opponent. Also shows A2C with the exact solver's values as a frozen critic.

Win rate against the best response (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 0%; the learners win 0%.



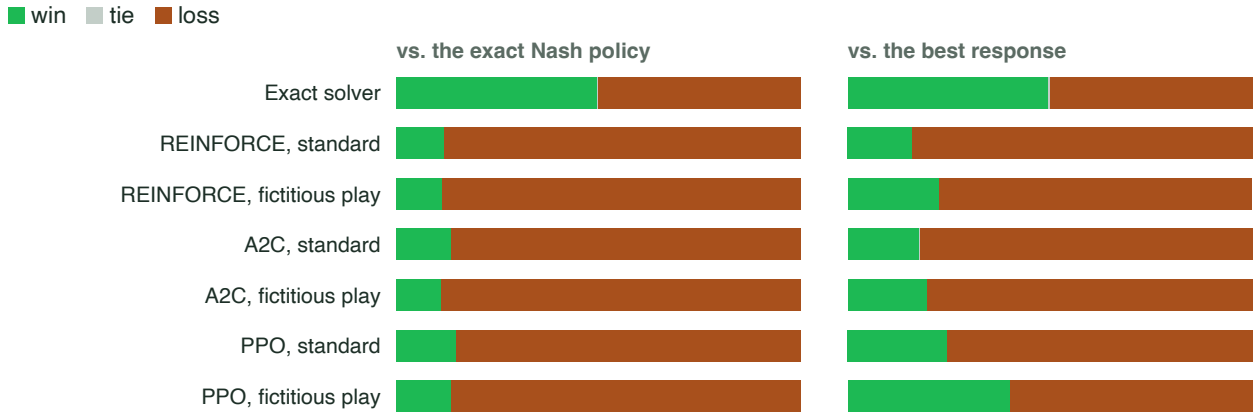
Wins, ties and losses against each opponent. Against the exact Nash policy the standard learners tie 34% to 100% of games and the fictitious-play learners 0% to 1%. A2C with the exact critic ties the exact Nash policy in 80% of games, against 45% for A2C with a learned critic.



learner	vs. a random player	vs. the exact Nash policy	vs. the best response
Exact solver	**91%** tie 9% · loss 0%	**0%** tie 100% · loss 0%	**0%** tie 100% · loss 0%
REINFORCE, standard	**79%** tie 20% · loss 1%	**0%** tie 34% · loss 66%	**0%** tie 0% · loss 100%
REINFORCE, fictitious play	**100%** tie 0% · loss 0%	**0%** tie 0% · loss 100%	**0%** tie 0% · loss 100%
A2C, standard	**64%** tie 35% · loss 2%	**0%** tie 45% · loss 55%	**0%** tie 0% · loss 100%
A2C, fictitious play	**100%** tie 0% · loss 0%	**0%** tie 0% · loss 100%	**0%** tie 0% · loss 100%
PPO, standard	**53%** tie 43% · loss 5%	**0%** tie 100% · loss 0%	**0%** tie 0% · loss 100%
PPO, fictitious play	**95%** tie 4% · loss 0%	**0%** tie 0% · loss 100%	**0%** tie 0% · loss 100%
A2C, exact critic, standard	**73%** tie 27% · loss 0%	**0%** tie 80% · loss 20%	**0%** tie 0% · loss 100%
A2C, exact critic, fictitious play	**97%** tie 2% · loss 0%	**0%** tie 1% · loss 99%	**0%** tie 0% · loss 100%

4. Mixed states of the random board

At 94 states the exact equilibrium has to randomize. Each learner starts 20 games from every one of them.



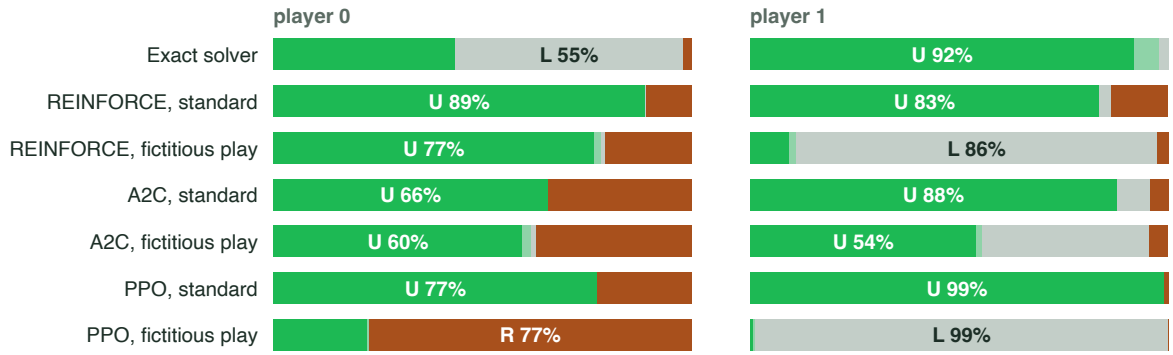
learner	vs. the exact Nash policy	vs. the best response	distance from the exact mix	player 0 mixes
Exact solver	**50%** tie 0% · loss 50%	**50%** tie 0% · loss 50%	0	–
REINFORCE, standard	**12%** tie 0% · loss 88%	**16%** tie 0% · loss 84%	0.64	72%
REINFORCE, fictitious play	**11%** tie 0% · loss 89%	**23%** tie 0% · loss 77%	0.65	98%
A2C, standard	**14%** tie 0% · loss 86%	**18%** tie 0% · loss 82%	0.59	71%
A2C, fictitious play	**11%** tie 0% · loss 89%	**20%** tie 0% · loss 80%	0.65	100%
PPO, standard	**15%** tie 0% · loss 85%	**25%** tie 0% · loss 75%	0.67	29%
PPO, fictitious play	**14%** tie 0% · loss 86%	**40%** tie 0% · loss 60%	0.64	72%

The distance from the exact mix is 0.59 to 0.67 for every learner. Player 0 mixes (puts under 90% on its top move) at 29% to 100% of these states, though not at the states or in the proportions the exact solution does; against the exact Nash policy the learners win 11% to 15% (exact solver 50%).

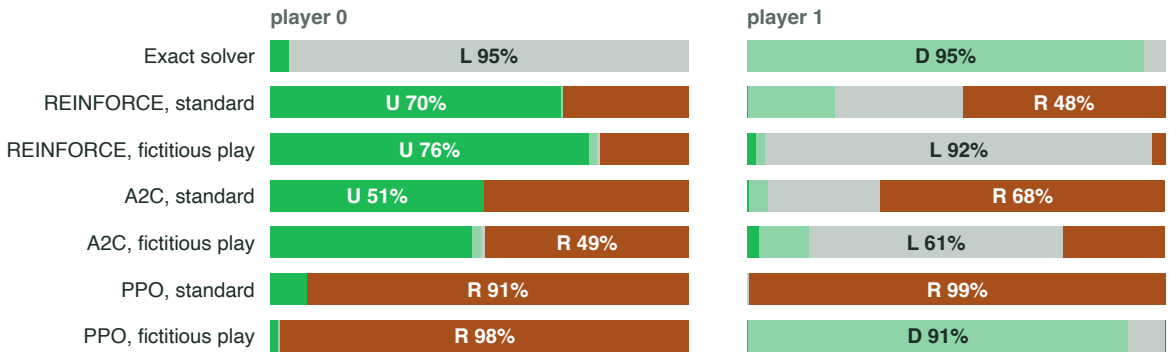
Three of these states in full



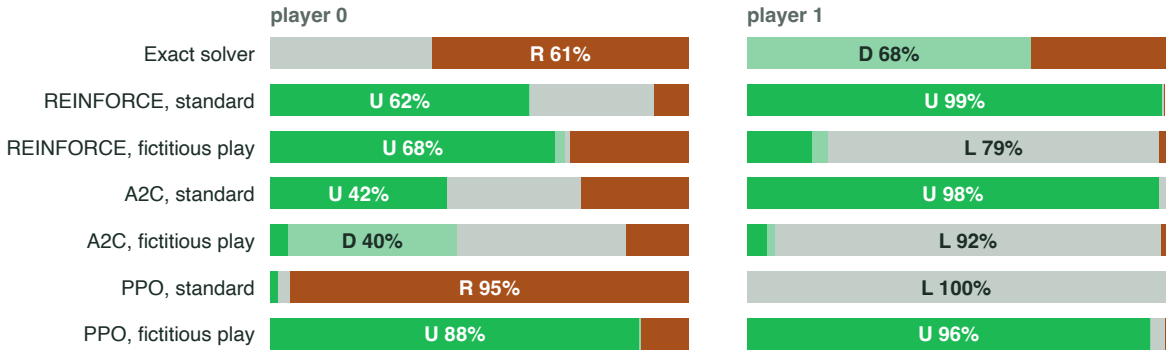
Mixed state 1 (three-way mix): player 0 at (0, 0), player 1 at (2, 0), player 0 has the ball



Mixed state 2 (two-way mix): player 0 at (0, 0), player 1 at (1, 1), player 0 has the ball



Mixed state 3 (two-way mix): player 0 at (0, 1), player 1 at (1, 0), player 1 has the ball



Mixed state 1 (three-way mix): player 0 at (0, 0), player 1 at (2, 0), player 0 has the ball

policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	43%	0%	**55%**	2%	**92%**	6%	2%	0%
REINFORCE, standard	**89%**	0%	0%	11%	**83%**	0%	3%	14%
REINFORCE, fictitious play	**77%**	2%	1%	21%	9%	2%	**86%**	3%
A2C, standard	**66%**	0%	0%	34%	**88%**	0%	8%	4%
A2C, fictitious play	**60%**	2%	1%	37%	**54%**	1%	40%	4%
PPO, standard	**77%**	0%	0%	23%	**99%**	0%	0%	1%
PPO, fictitious play	22%	0%	0%	**77%**	1%	1%	**99%**	0%

Mixed state 2 (two-way mix): player 0 at (0, 0), player 1 at (1, 1), player 0 has the ball

policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	5%	0%	**95%**	0%	0%	**95%**	5%	0%
REINFORCE, standard	**70%**	0%	0%	30%	0%	21%	31%	**48%**
REINFORCE, fictitious play	**76%**	2%	1%	21%	2%	2%	**92%**	3%
A2C, standard	**51%**	0%	0%	49%	0%	5%	27%	**68%**
A2C, fictitious play	48%	2%	1%	**49%**	3%	12%	**61%**	24%
PPO, standard	9%	0%	0%	**91%**	0%	0%	0%	**99%**
PPO, fictitious play	2%	0%	0%	**98%**	0%	**91%**	9%	0%

Mixed state 3 (two-way mix): player 0 at (0, 1), player 1 at (1, 0), player 1 has the ball

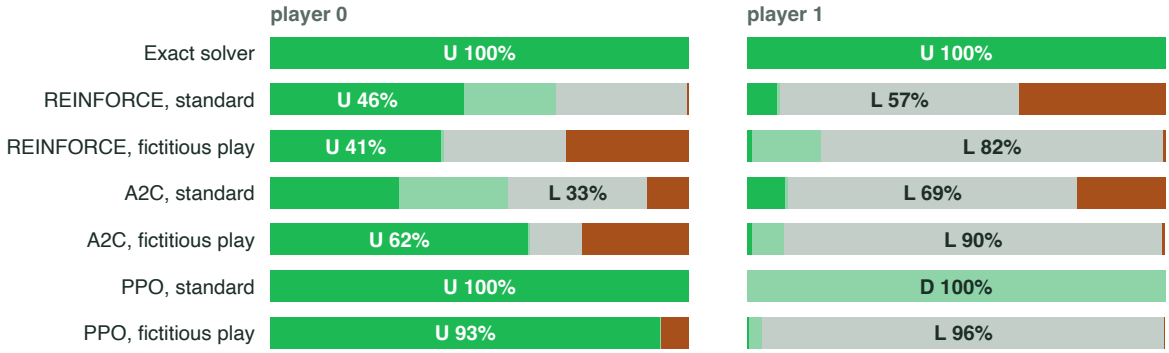
policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	0%	0%	39%	**61%**	0%	**68%**	0%	32%
REINFORCE, standard	**62%**	0%	29%	8%	**99%**	0%	1%	0%
REINFORCE, fictitious play	**68%**	2%	1%	28%	15%	4%	**79%**	2%
A2C, standard	**42%**	0%	32%	26%	**98%**	0%	2%	0%
A2C, fictitious play	4%	**40%**	40%	15%	5%	2%	**92%**	1%
PPO, standard	2%	0%	3%	**95%**	0%	0%	**100%**	0%
PPO, fictitious play	**88%**	0%	0%	11%	**96%**	0%	3%	0%

5. Action probabilities

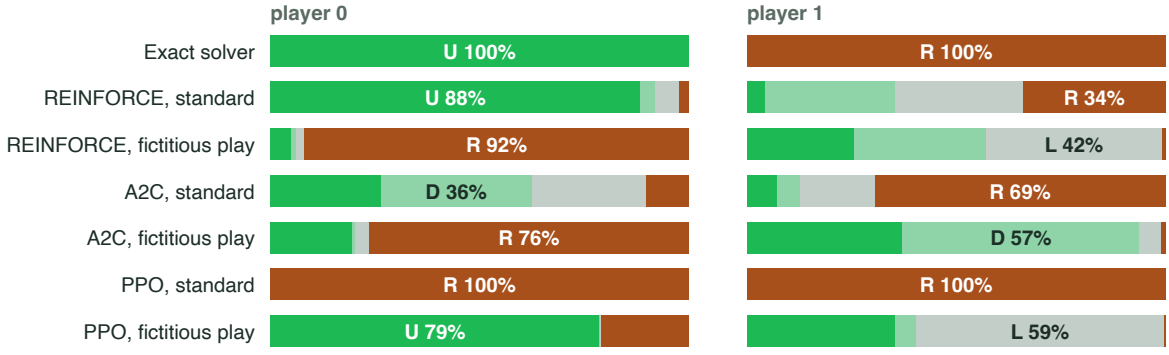
The probability each trained network gives to every move, at four fixed positions of the deterministic board (seed 0), next to the exact solver's move.



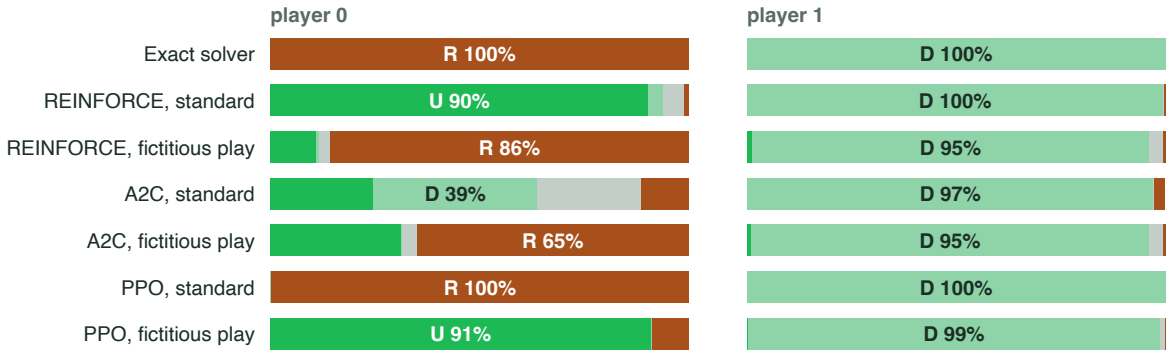
Kickoff: player 0 at (0, 1), player 1 at (6, 3), player 0 has the ball



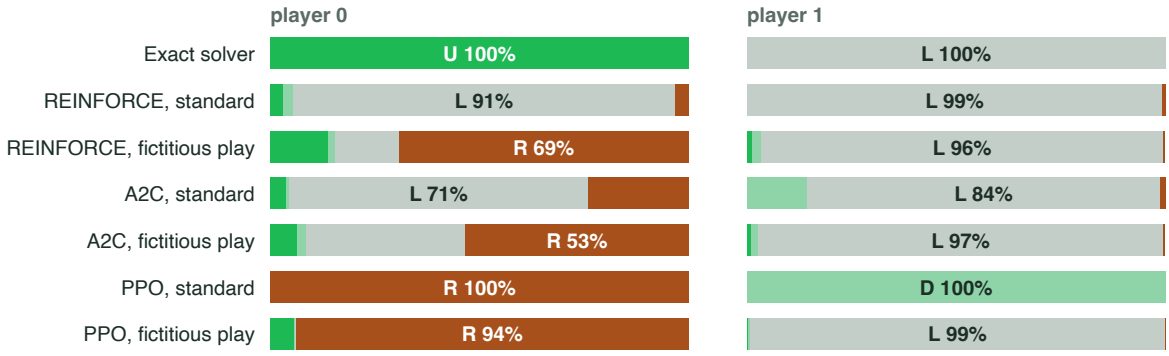
Players side by side in a goal row: player 0 at (3, 2), player 1 at (4, 2), player 0 has the ball



Carrier near the goal, defender far away: player 0 at (5, 1), player 1 at (6, 3), player 0 has the ball



The right player holds the ball, players stacked: player 0 at (3, 2), player 1 at (3, 3), player 1 has the ball



Kickoff: player 0 at (0, 1), player 1 at (6, 3), player 0 has the ball

policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	**100%**	0%	0%	0%	**100%**	0%	0%	0%
REINFORCE, standard	**46%**	22%	31%	0%	7%	1%	**57%**	35%
REINFORCE, fictitious play	**41%**	1%	29%	29%	1%	17%	**82%**	1%
A2C, standard	31%	26%	**33%**	10%	9%	1%	**69%**	21%
A2C, fictitious play	**62%**	0%	13%	25%	1%	8%	**90%**	1%
PPO, standard	**100%**	0%	0%	0%	0%	**100%**	0%	0%
PPO, fictitious play	**93%**	0%	0%	7%	0%	3%	**96%**	0%

Players side by side in a goal row: player 0 at (3, 2), player 1 at (4, 2), player 0 has the ball

policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	**100%**	0%	0%	0%	0%	0%	0%	**100%**
REINFORCE, standard	**88%**	3%	6%	2%	4%	31%	31%	**34%**
REINFORCE, fictitious play	5%	1%	2%	**92%**	25%	32%	**42%**	1%
A2C, standard	26%	**36%**	28%	10%	7%	5%	18%	**69%**
A2C, fictitious play	20%	1%	3%	**76%**	37%	**57%**	5%	1%
PPO, standard	0%	0%	0%	**100%**	0%	0%	0%	**100%**
PPO, fictitious play	**79%**	0%	0%	21%	35%	5%	**59%**	0%

Carrier near the goal, defender far away: player 0 at (5, 1), player 1 at (6, 3), player 0 has the ball

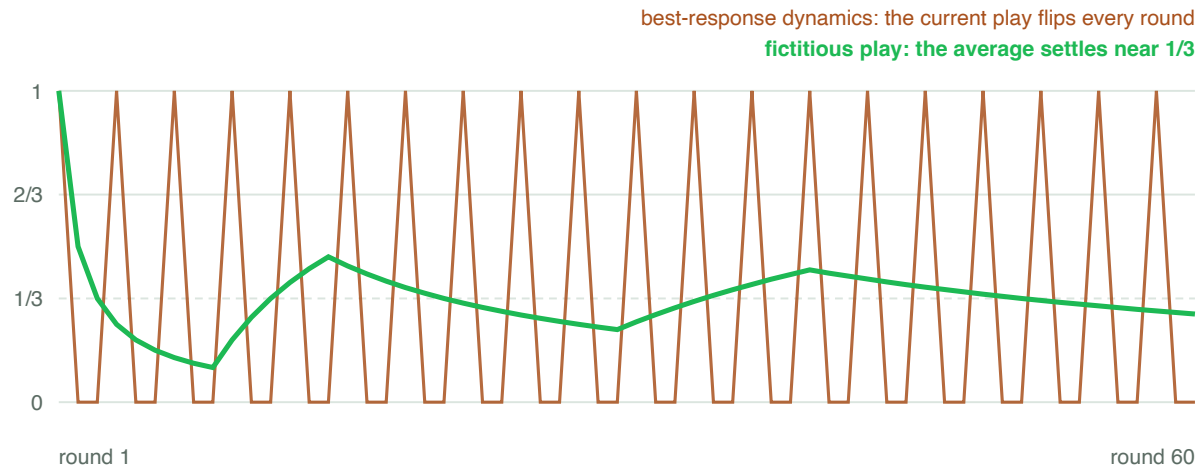
policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	0%	0%	0%	**100%**	0%	**100%**	0%	0%
REINFORCE, standard	**90%**	4%	5%	1%	0%	**100%**	0%	0%
REINFORCE, fictitious play	11%	1%	3%	**86%**	1%	**95%**	3%	1%
A2C, standard	25%	**39%**	25%	11%	0%	**97%**	0%	3%
A2C, fictitious play	31%	0%	4%	**65%**	1%	**95%**	3%	1%
PPO, standard	0%	0%	0%	**100%**	0%	**100%**	0%	0%
PPO, fictitious play	**91%**	0%	0%	9%	0%	**99%**	1%	0%

The right player holds the ball, players stacked: player 0 at (3, 2), player 1 at (3, 3), player 1 has the ball

policy	player 0				player 1			
	U	D	L	R	U	D	L	R
Exact solver	**100%**	0%	0%	0%	0%	0%	**100%**	0%
REINFORCE, standard	3%	3%	**91%**	3%	0%	0%	**99%**	1%
REINFORCE, fictitious play	14%	2%	15%	**69%**	1%	2%	**96%**	0%
A2C, standard	4%	1%	**71%**	24%	0%	14%	**84%**	1%
A2C, fictitious play	6%	2%	38%	**53%**	1%	2%	**97%**	0%
PPO, standard	0%	0%	0%	**100%**	0%	**100%**	0%	0%
PPO, fictitious play	6%	0%	0%	**94%**	0%	0%	**99%**	0%

6. Why fictitious play

On rock-paper-scissors, answering the opponent's latest move makes the play flip between rock, paper and scissors for ever. Answering the average of everything it has played converges to one third each.

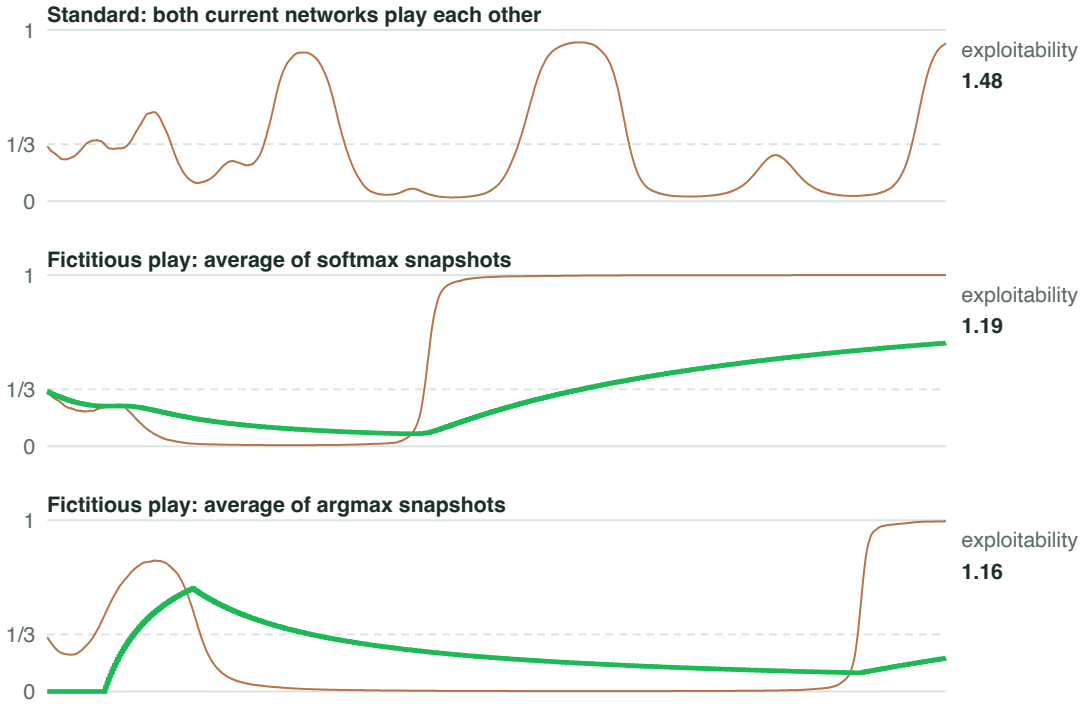


The same test with neural networks

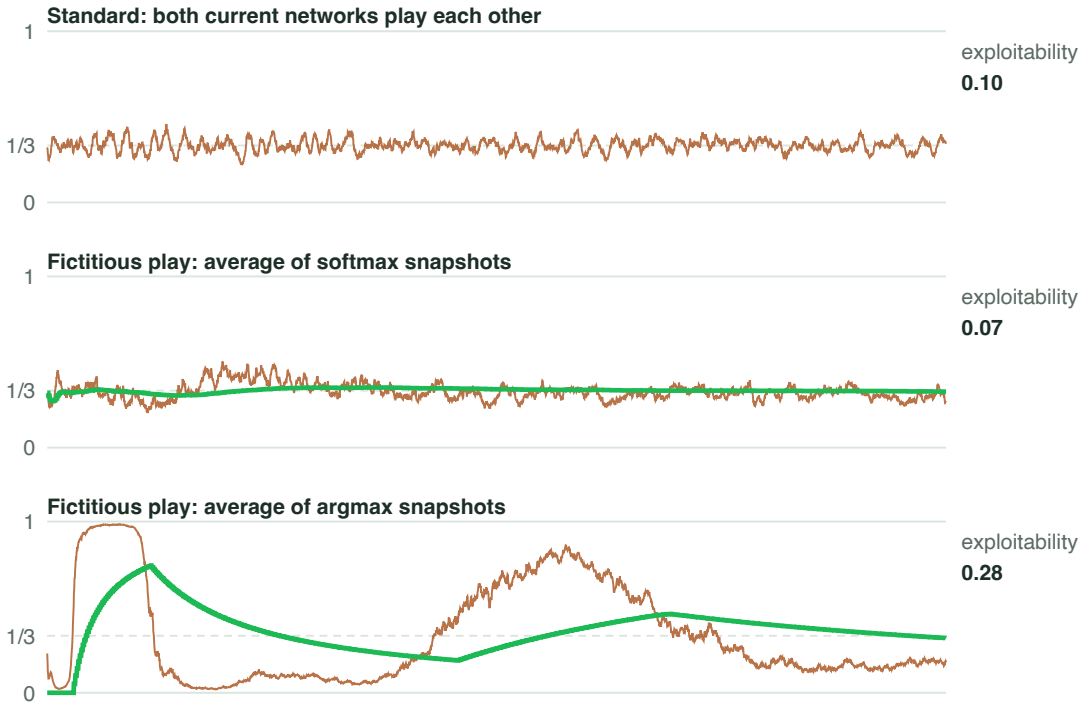
Each player is a small network with a softmax output, trained by REINFORCE on sampled games (3,000 iterations, 64 games each, 3 seeds; the plots show seed 0). **Standard** trains the two current networks against each other. **Fictitious play** trains each player against a snapshot of the other drawn from all its past snapshots and reports the average of a player's snapshots; **argmax** first turns each snapshot into the pure policy that plays its most likely move, so the average is how often the snapshots play each move. Orange: the current network's share of rock; green: the aggregate. Exploitability (right) is what a perfect attacker could gain against the two networks' policies, summed over both players: 0 at one third each, 2 at worst; it is the mean over seeds and the last 300 iterations. With the soccer settings the standard networks cycle (exploitability 1.48), and neither form of fictitious play reaches one third (softmax 1.19, argmax 1.16). With the larger entropy

bonus all three settle close to the equilibrium (standard 0.10, softmax 0.07, argmax 0.28): this setting damps the cycling, so the standard networks settle too (a stabilization effect, not evidence of convergence to the unregularized equilibrium).

Soccer settings: Adam at 0.001, entropy bonus 0.01



Larger entropy bonus: plain gradient steps at 0.1, entropy bonus 0.2



How long to train each best response

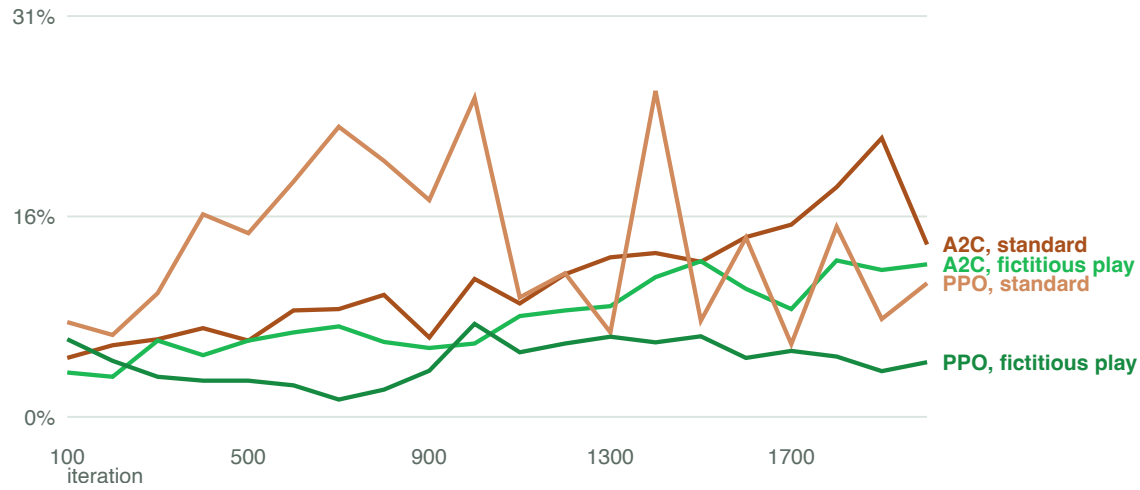
Same networks and settings, 8,000 iterations, 3 seeds; each snapshot is taken after 5, 50 or 200 iterations of training. Exploitability of the average of snapshots (mean of the last 300 iterations; 0 is one third each):

time between snapshots	average of softmax snapshots	average of argmax snapshots
5 iterations	**1.23** 1.10 to 1.36	**1.25** 0.92 to 1.44
50 iterations	**1.28** 1.09 to 1.50	**1.29** 1.21 to 1.36
200 iterations	**1.17** 0.99 to 1.29	**1.04** 1.02 to 1.07

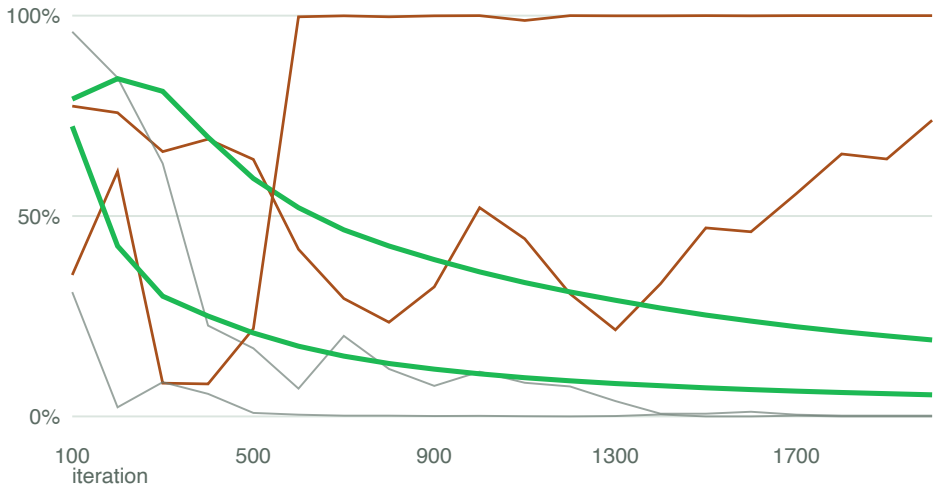
7. Does training converge?

Best-response dynamics (standard training) can cycle; fictitious play averages the past. Every 100 iterations the policy each learner would report is scored against the exact best response (mean of 3 seeds), and the probability it gives to the exact solver's kickoff move is recorded for seed 0: the current network (thin) and, for fictitious play, the average (thick). Orange: standard training's current network. Grey: the current network under fictitious play. Green: fictitious play's average. A2C and PPO are drawn together in each chart.

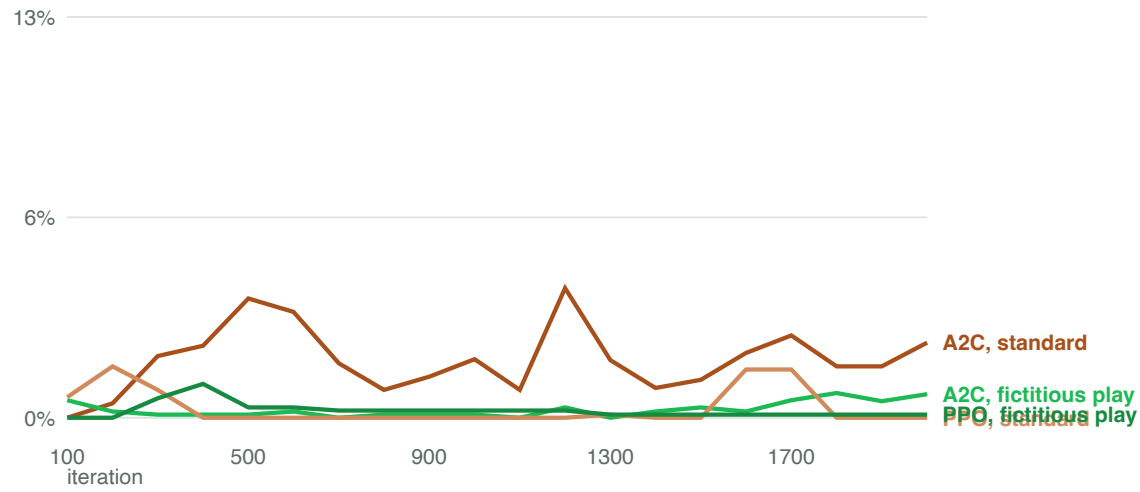
Random move-order board: win rate against the best response



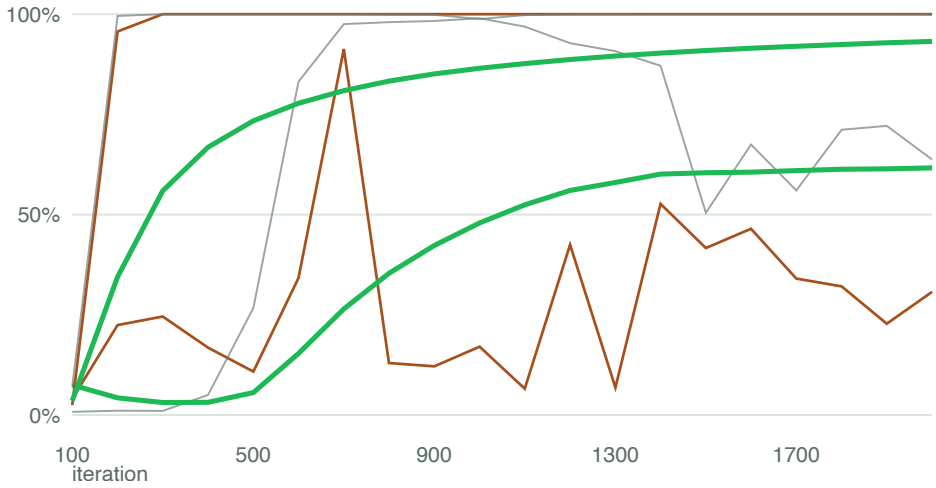
Random move-order board: probability of the exact solver's kickoff move (R; exact solver 100%)



Deterministic board: win rate against the best response



Deterministic board: probability of the exact solver's kickoff move (U; exact solver 100%)

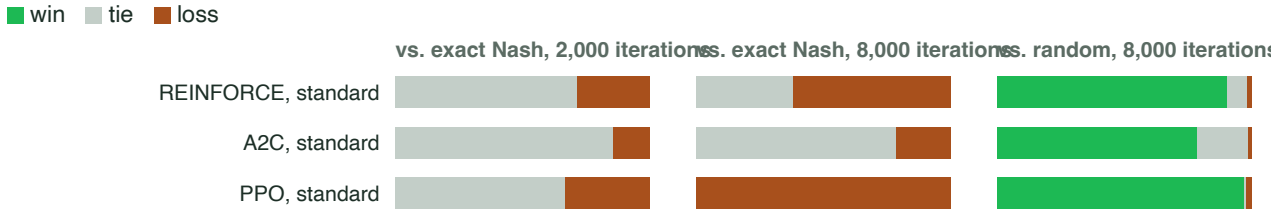


Random board, second half of training: the win rate against the best response stays within A2C standard 9% to 22%; A2C fictitious play 8% to 12%; PPO standard 6% to 25%; PPO fictitious play 4% to 6%. The exact solver wins 67% when it starts with the ball.

Deterministic board, second half of training: the win rate against the best response stays within A2C standard 1% to 4%; A2C fictitious play 0% to 1%; PPO standard 0% to 2%; PPO fictitious play 0%. The exact solver wins 0% when it starts with the ball.

8. More training

Standard training at 8,000 instead of 2,000 iterations (two seeds each), deterministic board.

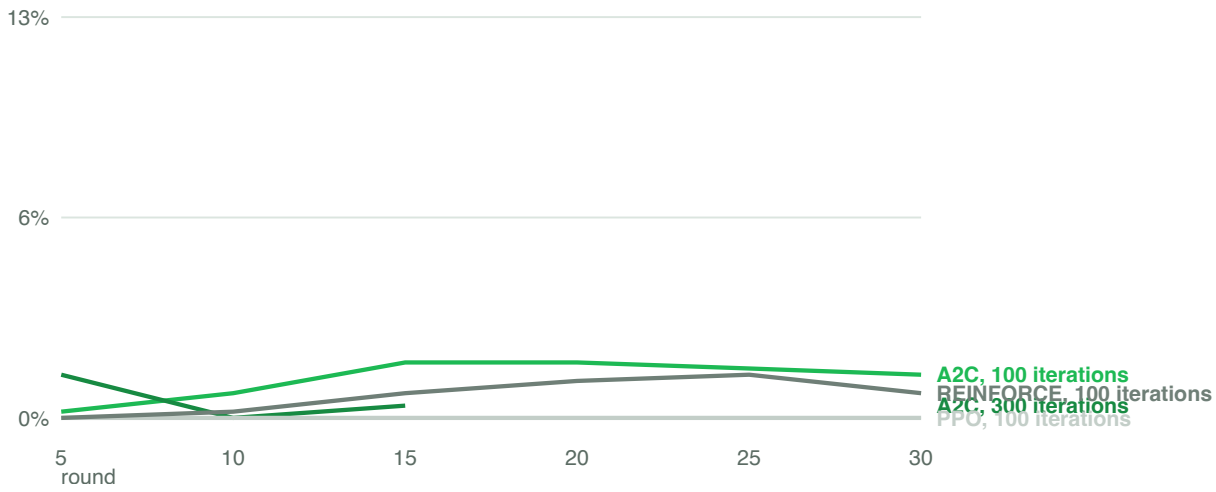


learner	vs. exact Nash, 2,000 iterations	vs. exact Nash, 8,000 iterations	vs. random, 8,000 iterations
REINFORCE, standard	**0%** tie 71% · loss 29%	**0%** tie 38% · loss 62%	**91%** tie 8% · loss 2%
A2C, standard	**0%** tie 85% · loss 15%	**0%** tie 79% · loss 21%	**79%** tie 20% · loss 1%
PPO, standard	**0%** tie 67% · loss 33%	**0%** tie 0% · loss 100%	**97%** tie 0% · loss 2%

At 8,000 iterations the learners win 0% of games against the exact Nash policy (at 2,000 iterations 0%) and 79% to 97% against a random player; more standard training does not close the gap to the exact solver.

9. Fictitious play with long best responses

Each round a player trains for 100 (or 300) iterations against the average of its opponent's earlier best responses, then adds the result to its history. Win rate against the best response at each checkpoint round (one seed per run); every line ends below 2%.

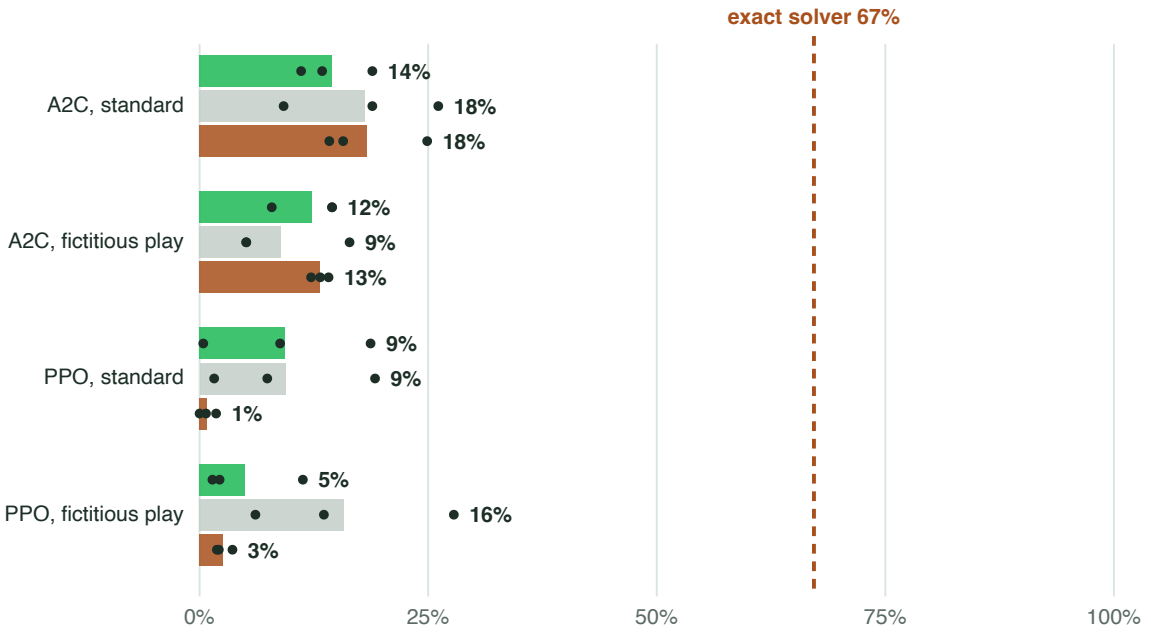


run	round 5	round 10	round 15	round 20	round 25	round 30
A2C, 100 iterations per best response	0%	1%	2%	2%	2%	1%
A2C, 300 iterations per best response	1%	0%	0%			
PPO, 100 iterations per best response	0%	0%	0%	0%	0%	0%
REINFORCE, 100 iterations per best response	0%	0%	1%	1%	1%	1%

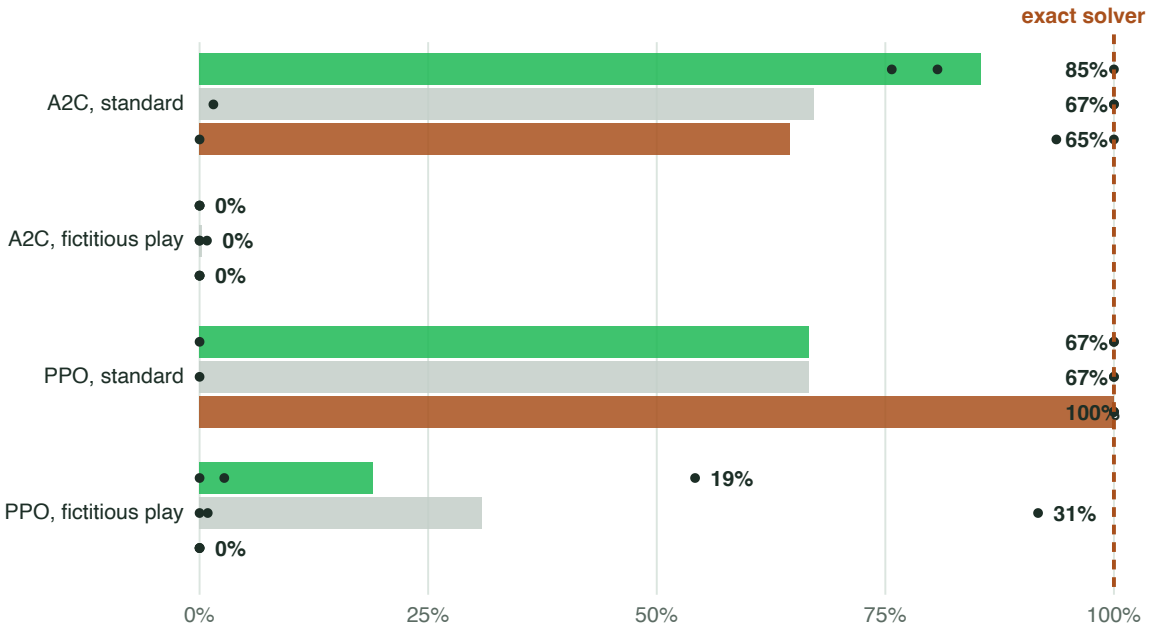
10. Network sharing and trimming

A2C and PPO, three seeds each, on all three boards. **Shared network**: both players' policies are two output slices of one network, so each player's update also moves the other. **Last 10 steps left out**: the final 10 of the 100 steps of every episode are not used in the loss, though they still count in the returns of earlier steps. Bars are the mean over seeds; dots are single seeds. ■ separate networks, ■ one shared network, ■ last 10 steps left out.

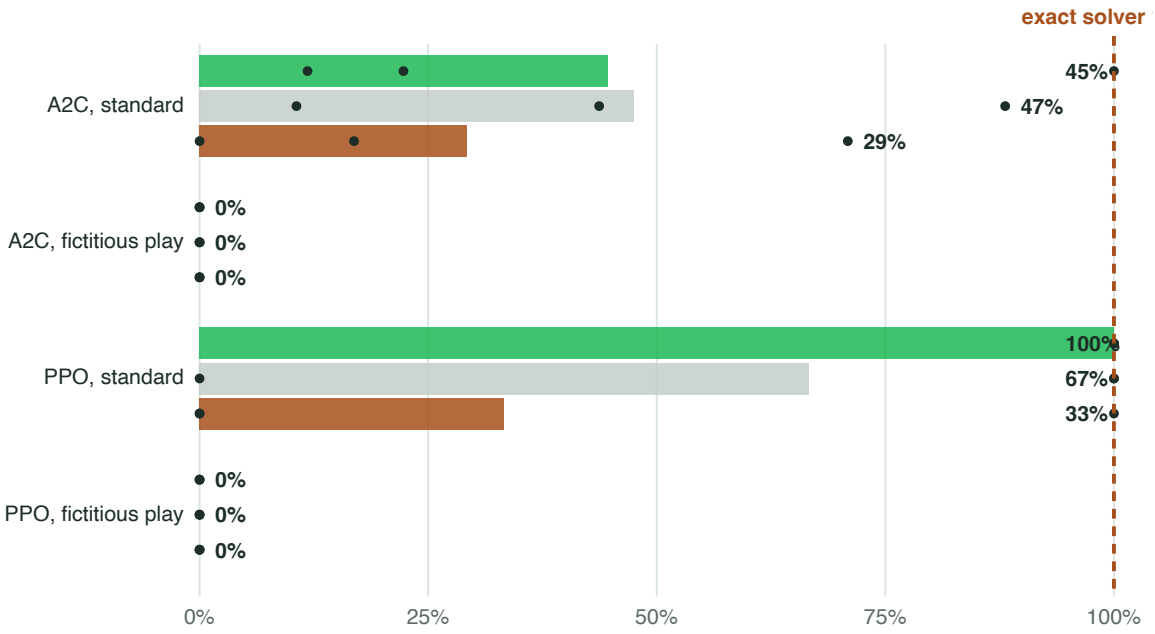
Random move-order board: win rate against the best response



Deterministic board: games tied with the exact Nash policy



Continuing game: games tied with the exact Nash policy



Random board: against separate networks, a shared network changes the win rate by +4, –3, +0, +11 points (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 22 points. Leaving out the last 10 steps changes A2C by +4 and +1 points, but PPO with standard training stops scoring: it wins 19% of games against a random player, against 92% before. No variant gets close to the exact solver.

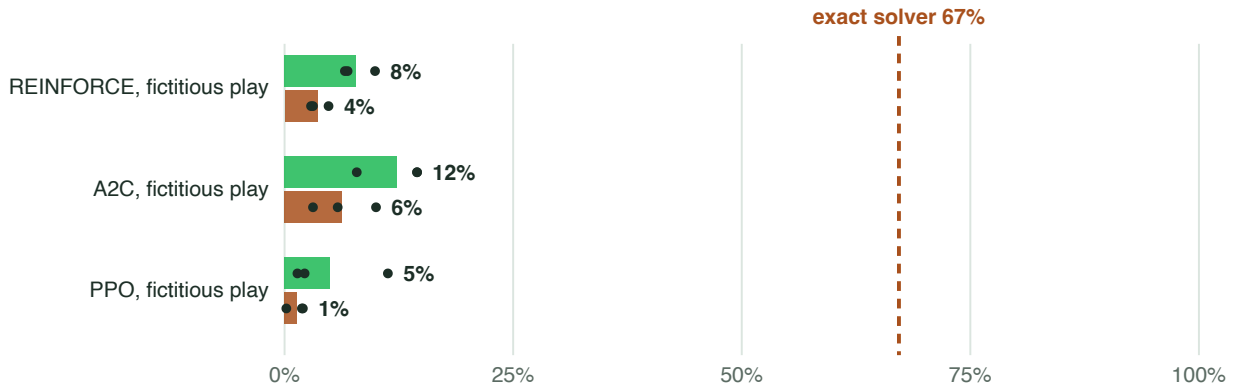
Deterministic board: no variant wins more than 2% of games against the best response, as for the baseline (the exact solver wins 0%, since it ties itself). The share of games tied with the exact Nash policy changes by –18, +0, +0, +12 points with a shared network and by –21, +0, +33, –19 points with the last 10 steps left out (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 100 points. A tie does not mean equilibrium play: PPO with standard training ties all games once the last 10 steps are left out, but wins only 36% of games against a random player, against 61% before.

Continuing game: no variant wins any games against the best response, as for the baseline. The share of games tied with the exact Nash policy changes by +3, +0, –33, +0 points with a shared network and by –15, +0, –67, +0 points with the last 10 steps left out (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 100 points, so three seeds cannot settle changes of this size.

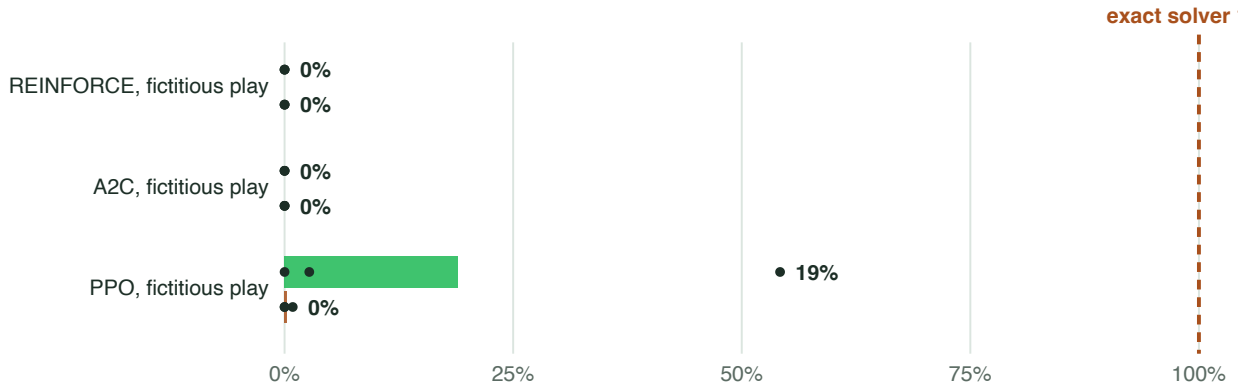
11. Averaging pure policies

Neural networks cannot output a mixed strategy except through their softmax, so each snapshot is made a pure policy (the move with the highest probability), those are averaged, and each player best-responds to that average. This is fictitious play in its original form: each snapshot is a pure best response, and the mix is how often each move was played. Both the opponent that a player trains against and the policy it reports are averages of pure snapshots. Bars are the mean over 3 seeds and dots single seeds; ■ average of softmax snapshots (the fictitious play used elsewhere), ■ average of argmax snapshots.

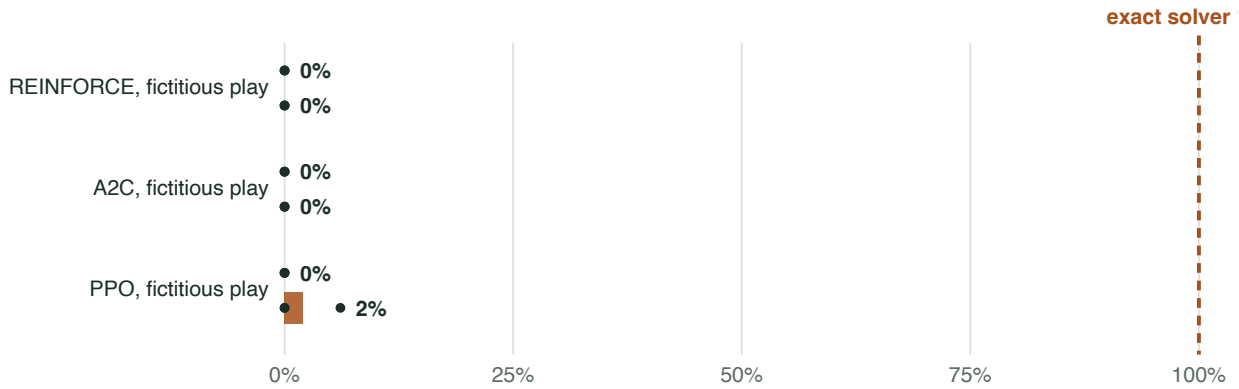
Random move-order board: win rate against the best response



Deterministic board: games tied with the exact Nash policy



Continuing game: games tied with the exact Nash policy



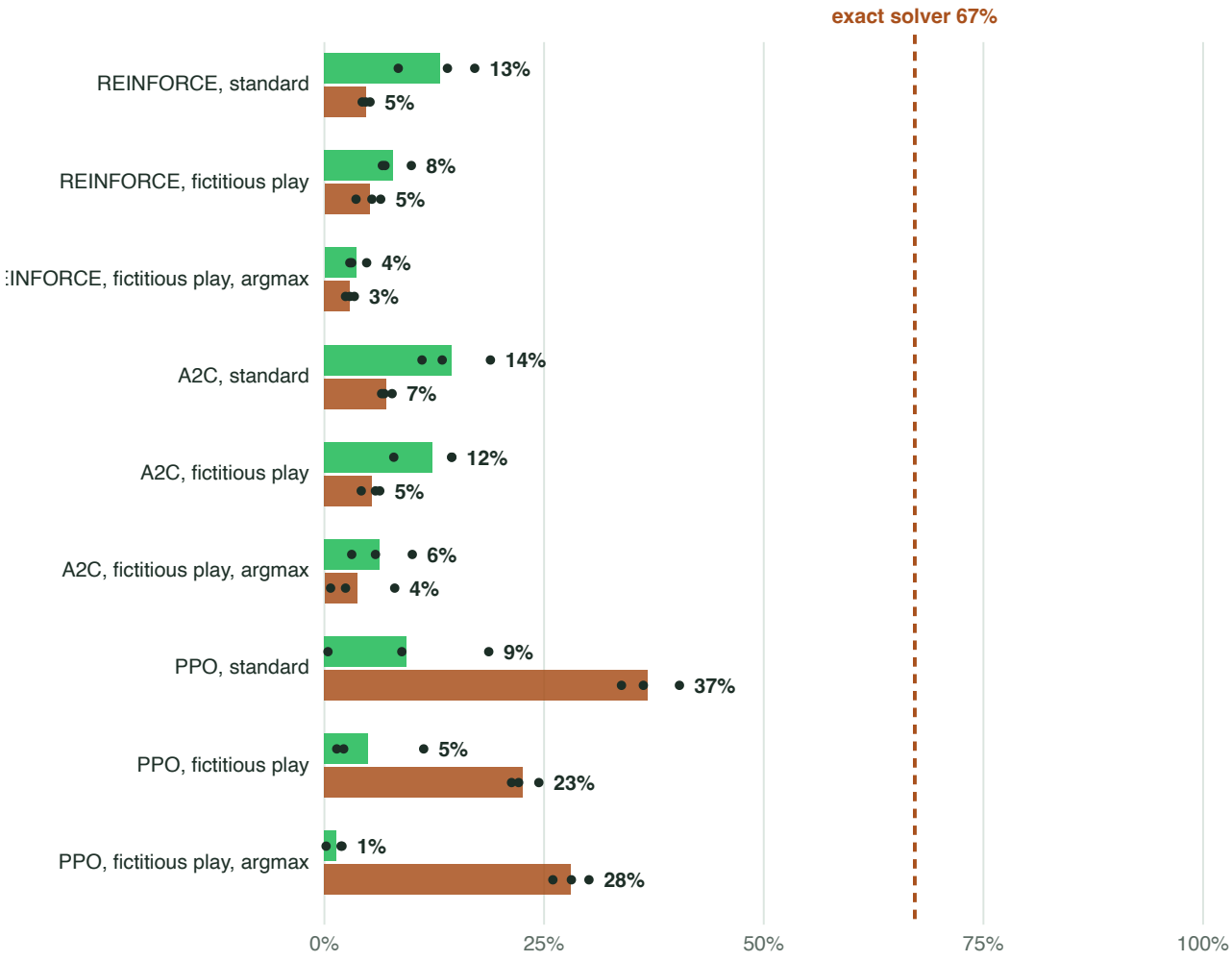
Random board: averaging argmax instead of softmax snapshots changes the win rate against the best response by –4, –6, –4 points (REINFORCE, A2C, PPO); single seeds differ by up to 10 points. Deterministic board: it changes the share of games tied with the exact Nash policy by +0, +0, –19 points; single seeds differ by up to 54 points. Continuing game: the share tied with the exact Nash policy changes by +0, +0, +2 points; single seeds differ by up to 6 points, and no argmax learner wins any games of games against the best response.

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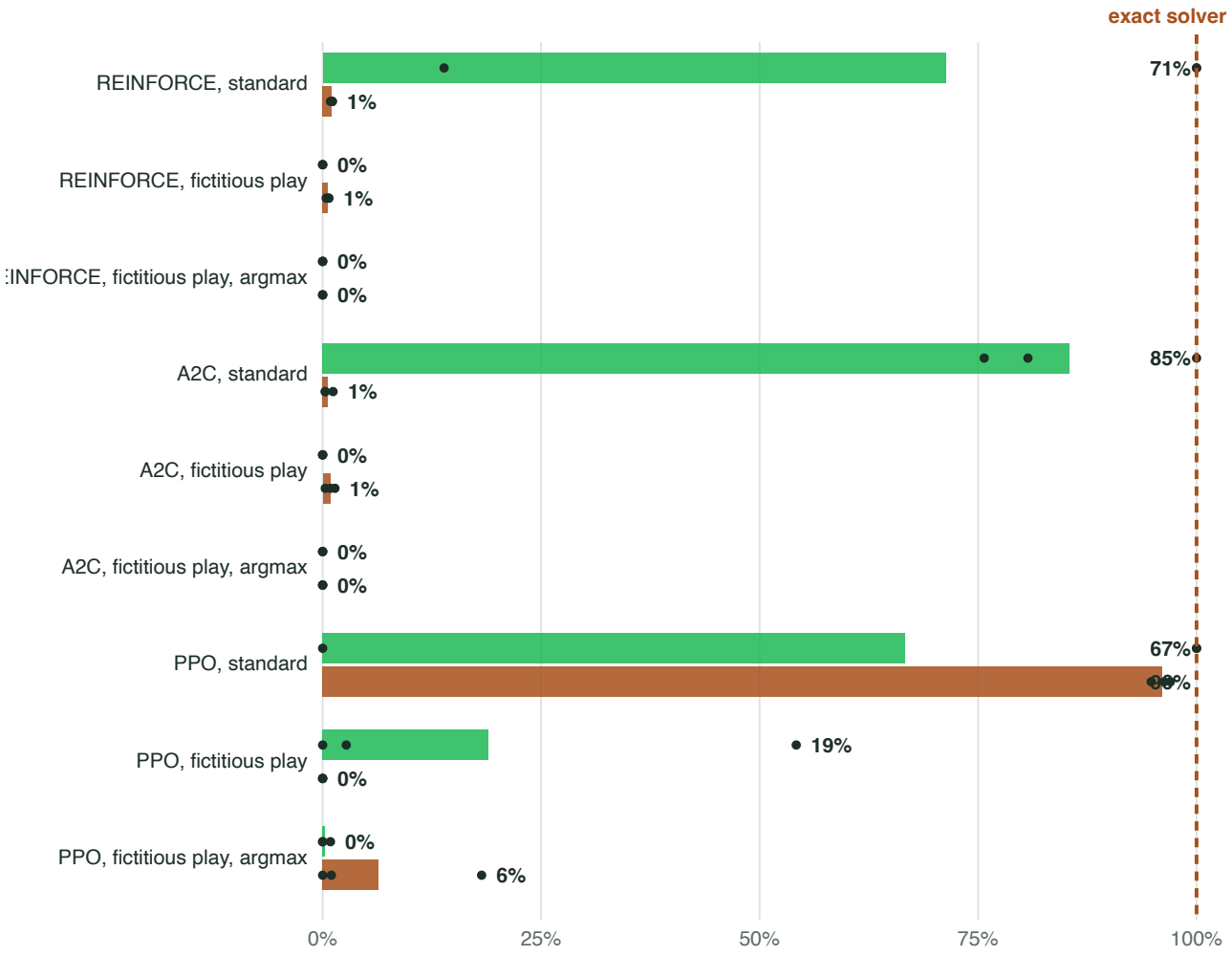
12. A larger entropy bonus

On rock-paper-scissors a larger entropy bonus made the networks settle, but entropy regularization changes the optimization problem, so this is a stabilization heuristic reported next to the original setting, not a way to recover the unregularized equilibrium. Here every learner (REINFORCE, A2C, PPO), every way of training (standard, fictitious play, fictitious play with argmax snapshots) and every board is trained again with the entropy bonus raised from 0.01 to 0.2; nothing else changes. Bars are the mean over 3 seeds and dots single seeds; ■ entropy bonus 0.01 (used elsewhere), ■ entropy bonus 0.2.

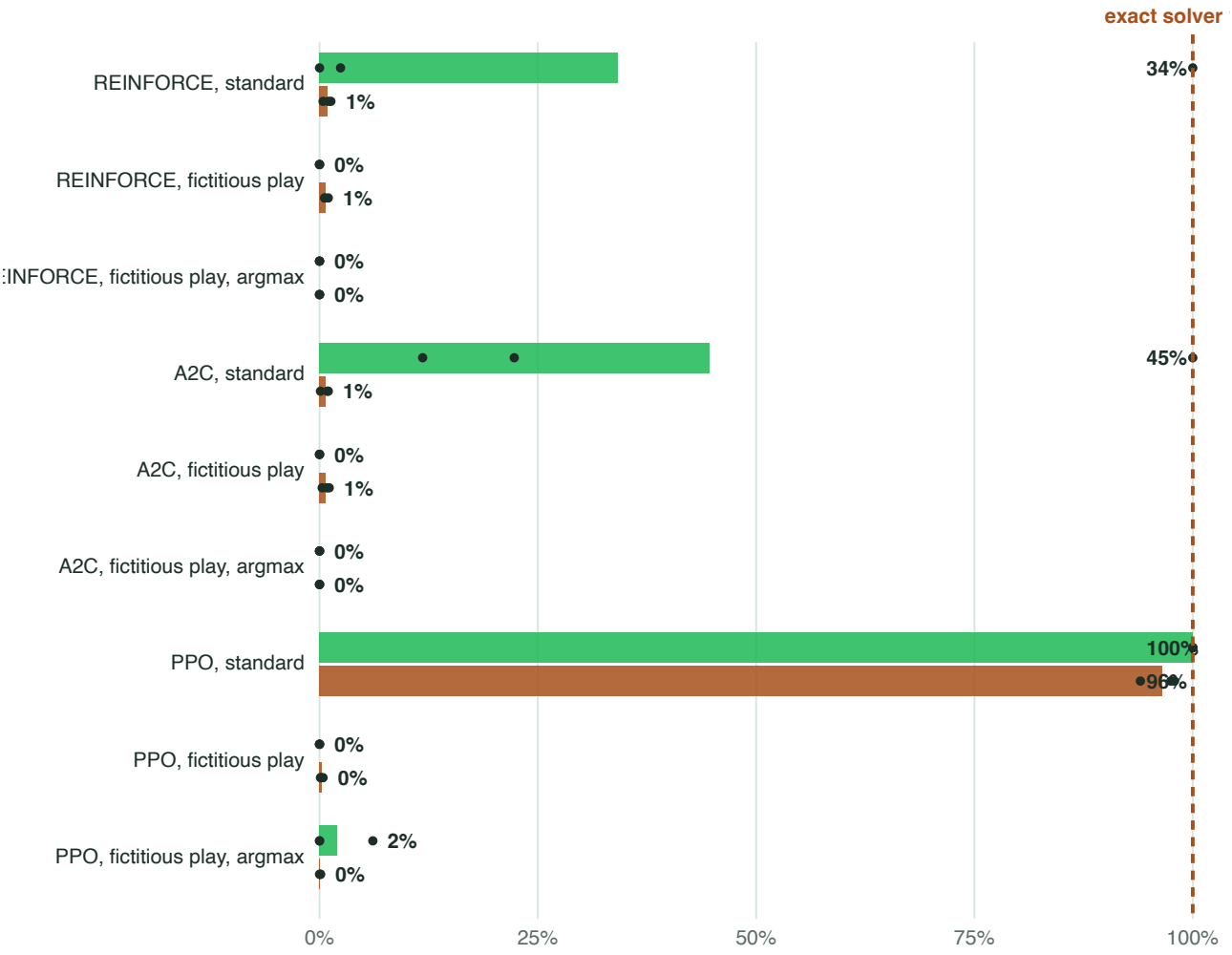
Random move-order board: win rate against the best response



Deterministic board: games tied with the exact Nash policy



Continuing game: games tied with the exact Nash policy



On the random board, averaged over the three ways of training, the larger bonus moves the win rate against the best response by REINFORCE −4, A2C −6, PPO +24 points; the exact solver wins 67% when it starts with the ball. This is an ablation and does not show convergence to the unregularized equilibrium.

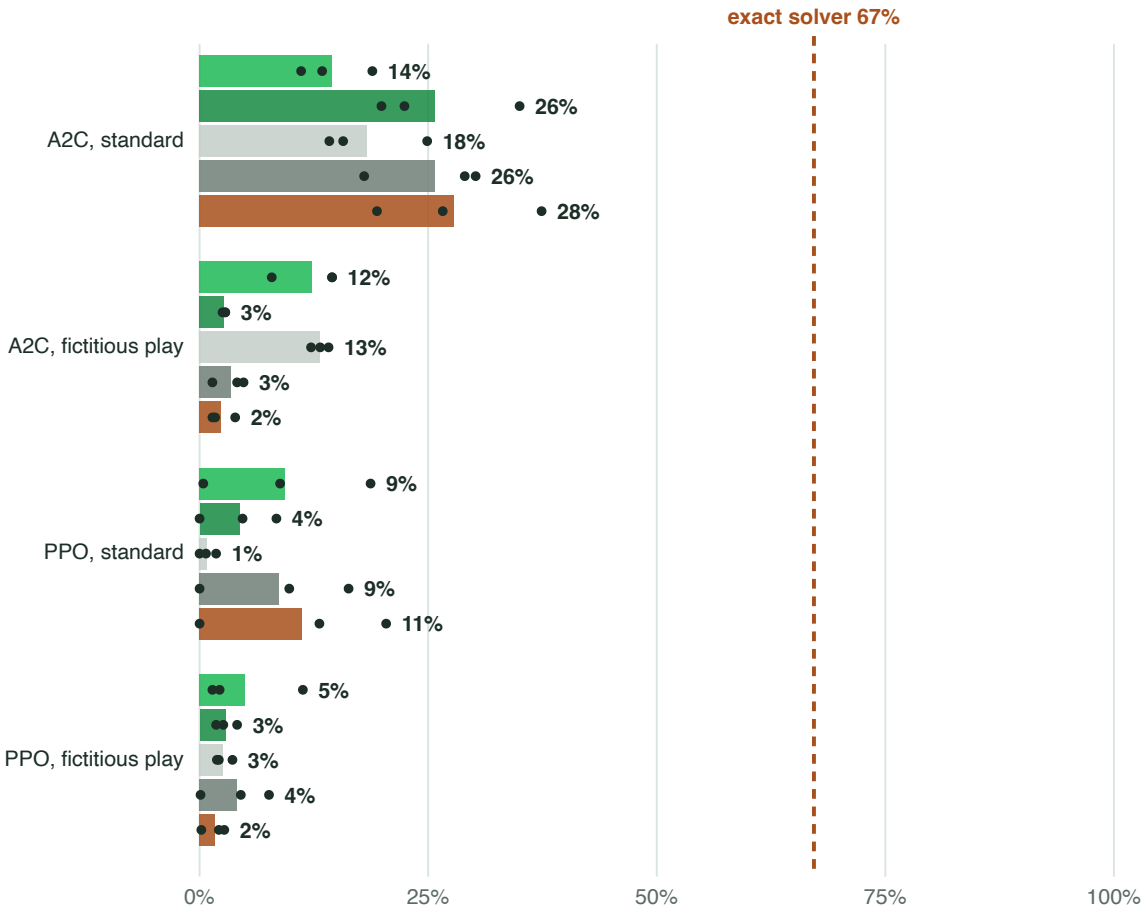
Random board: the larger bonus changes the win rate against the best response by −8, −3, −1, −7, −7, −3, +28, +18, +27 points; against a random player the win rate changes by +1, −8, +5, −2, −6, +4, +6, +4, +3 points (REINFORCE, A2C, PPO; standard, fictitious play, argmax). Deterministic board: the larger bonus changes the share tied with the exact Nash policy by −70, +1, +0, −85, +1, +0, +29, −19, +6 points; against a random player the win rate changes by +22, −2, +5, +19, −6, +4, +32, +11, +13 points (REINFORCE, A2C, PPO; standard, fictitious play, argmax). Continuing game: the larger bonus changes the share tied with the exact Nash policy by −33, +1, +0, −44, +1, +0, −4, +0, −2 points; against a random player the win rate changes by +6, −17, +6, +24, −16, +6, +43, +5, +11 points (REINFORCE, A2C, PPO; standard, fictitious play, argmax).

13. Training choices

Each comparison changes one thing from the usual setting (2,000 iterations of 64 episodes, all data used once, a snapshot every 5 iterations, all 100 steps in the loss) and keeps the rest. Bars are the mean over 3 seeds and dots single seeds; in the tables the mean is bold and the range of single seeds is beneath.

Leaving out the last steps: how many?

Random move-order board; the last N of the 100 steps of every episode are left out of the loss.

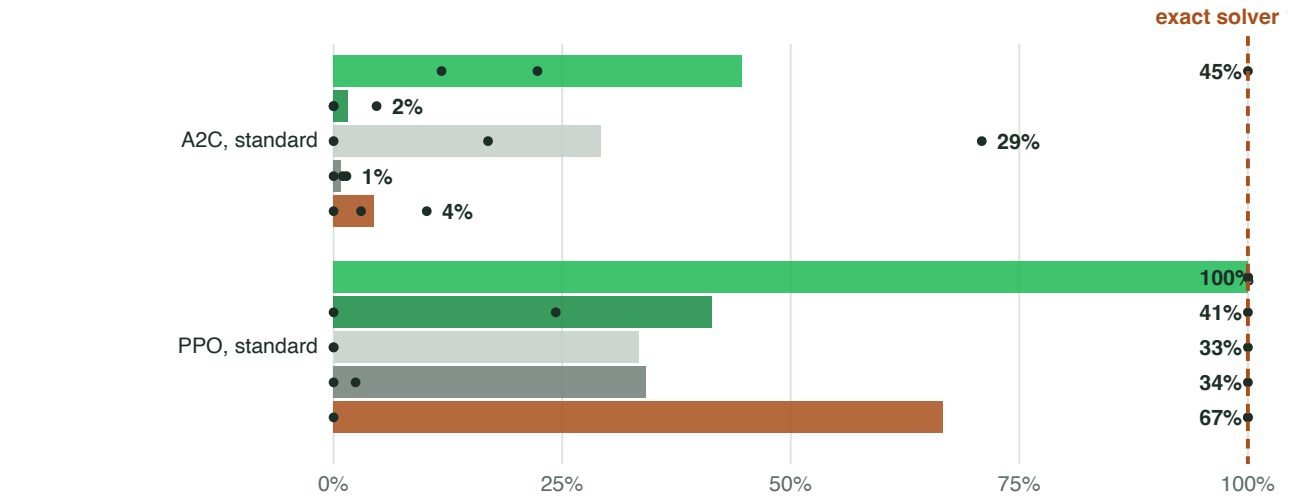


learner	all steps	last 5 left out	last 10 left out	last 20 left out	last 40 left out
A2C, standard	**14%** 11% to 19%	**26%** 20% to 35%	**18%** 14% to 25%	**26%** 18% to 30%	**28%** 19% to 37%
A2C, fictitious play	**12%** 8% to 14%	**3%** 2% to 3%	**13%** 12% to 14%	**3%** 1% to 5%	**2%** 1% to 4%
PPO, standard	**9%** 0% to 19%	**4%** 0% to 8%	**1%** 0% to 2%	**9%** 0% to 16%	**11%** 0% to 20%
PPO, fictitious play	**5%** 1% to 11%	**3%** 2% to 4%	**3%** 2% to 4%	**4%** 0% to 8%	**2%** 0% to 3%

Change in the win rate against the best response, in points, from "all steps" (rows in the order shown): last 5 left out: +11, −10, −5, −2; last 10 left out: +4, +1, −8, −2; last 20 left out: +11, −9, −1, −1; last 40 left out: +13, −10, +2, −3. Single seeds of one setting differ by up to 20 points.

Leaving out the last steps in the continuing game

Continuing game, standard training; share of games tied with the exact Nash policy.



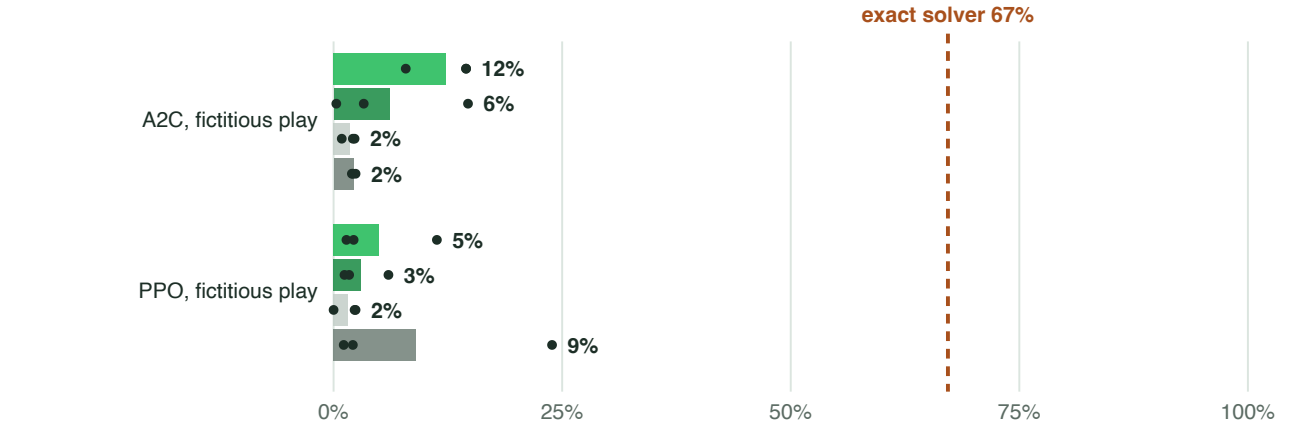
all steps · last 5 left out · last 10 left out · last 20 left out · last 40 left out

learner	all steps	last 5 left out	last 10 left out	last 20 left out	last 40 left out
A2C, standard	**45%** 12% to 100%	**2%** 0% to 5%	**29%** 0% to 71%	**1%** 0% to 1%	**4%** 0% to 10%
PPO, standard	**100%** 100%	**41%** 0% to 100%	**33%** 0% to 100%	**34%** 0% to 100%	**67%** 0% to 100%

Change in the share tied with the exact Nash policy, in points, from "all steps" (rows in the order shown): last 5 left out: –43, –59; last 10 left out: –15, –67; last 20 left out: –44, –66; last 40 left out: –40, –33. Single seeds of one setting differ by up to 100 points.

How long to train between snapshots

Random board, fictitious play, 2,000 iterations in total: the training budget of each best response.



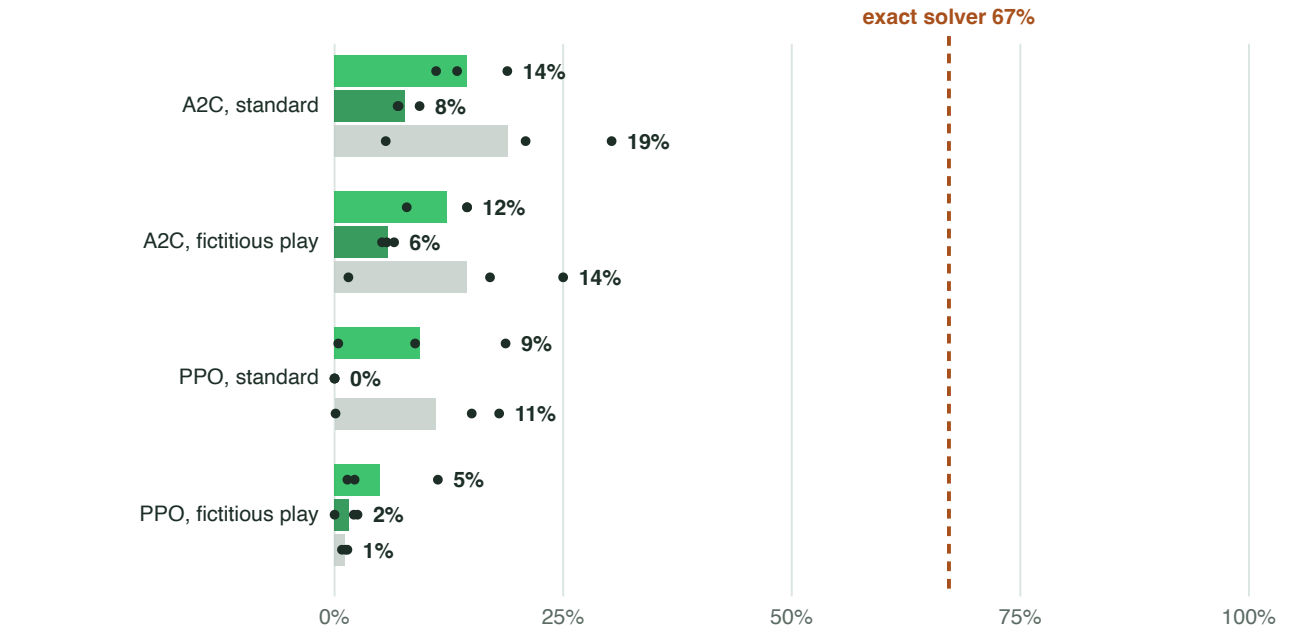
snapshot every 5 iterations (used elsewhere) · every iteration · every 25 · every 100

learner	snapshot every 5 iterations (used elsewhere)	every iteration	every 25	every 100
A2C, fictitious play	**12%** 8% to 14%	**6%** 0% to 15%	**2%** 1% to 2%	**2%** 2%
PPO, fictitious play	**5%** 1% to 11%	**3%** 1% to 6%	**2%** 0% to 2%	**9%** 1% to 24%

Change in the win rate against the best response, in points, from "snapshot every 5 iterations (used elsewhere)" (rows in the order shown): every iteration: –6, –2; every 25: –11, –3; every 100: –10, +4. Single seeds of one setting differ by up to 23 points.

How much data per iteration

Random board: 8, 64 or 256 episodes of 100 steps are simulated before each update.

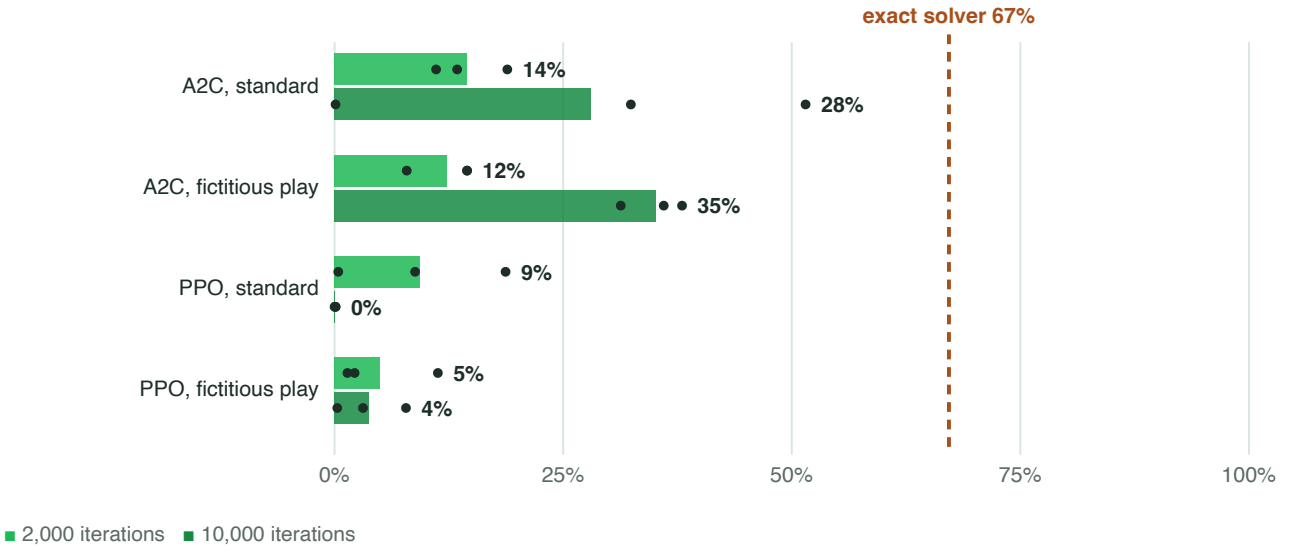


learner	64 episodes per iteration (used elsewhere)	8 episodes	256 episodes
A2C, standard	**14%** 11% to 19%	**8%** 7% to 9%	**19%** 6% to 30%
A2C, fictitious play	**12%** 8% to 14%	**6%** 5% to 6%	**14%** 2% to 25%
PPO, standard	**9%** 0% to 19%	**0%** 0%	**11%** 0% to 18%
PPO, fictitious play	**5%** 1% to 11%	**2%** 0% to 2%	**1%** 1%

Change in the win rate against the best response, in points, from "64 episodes per iteration (used elsewhere)" (rows in the order shown): 8 episodes: −7, −6, −9, −3; 256 episodes: +4, +2, +2, −4. Single seeds of one setting differ by up to 25 points.

Five times as many iterations

Random board: 10,000 iterations instead of 2,000.

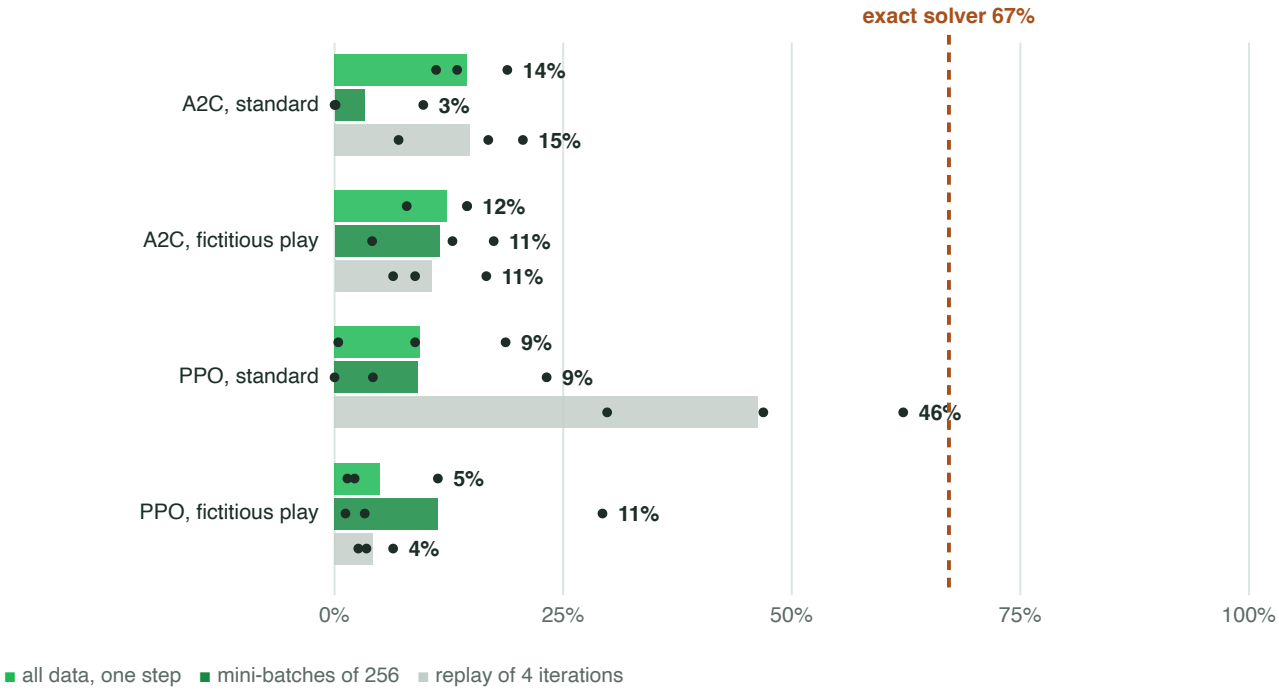


learner	2,000 iterations	10,000 iterations
A2C, standard	**14%** 11% to 19%	**28%** 0% to 52%
A2C, fictitious play	**12%** 8% to 14%	**35%** 31% to 38%
PPO, standard	**9%** 0% to 19%	**0%** 0%
PPO, fictitious play	**5%** 1% to 11%	**4%** 0% to 8%

Change in the win rate against the best response, in points, from "2,000 iterations" (rows in the order shown): 10,000 iterations: +14, +23, −9, −1. Single seeds of one setting differ by up to 51 points.

Mini-batches and replayed data

Random board. Mini-batch: many gradient steps per iteration, each on 256 random samples. Replay: each update also uses the data of the 3 previous iterations.

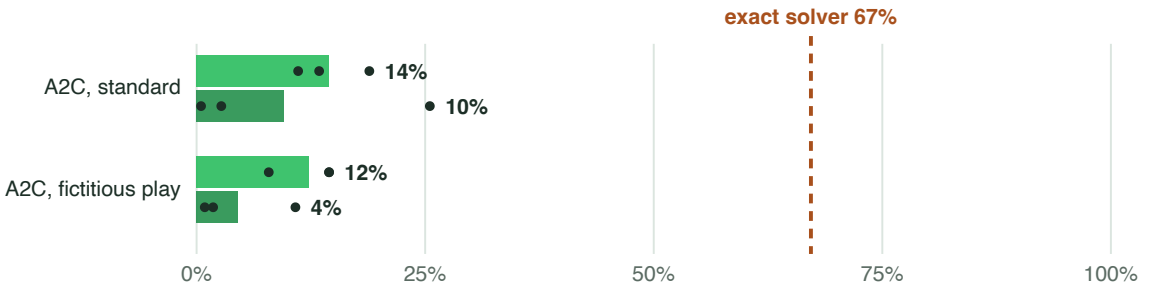


learner	all data, one step	mini-batches of 256	replay of 4 iterations
A2C, standard	**14%** 11% to 19%	**3%** 0% to 10%	**15%** 7% to 21%
A2C, fictitious play	**12%** 8% to 14%	**11%** 4% to 17%	**11%** 6% to 17%
PPO, standard	**9%** 0% to 19%	**9%** 0% to 23%	**46%** 30% to 62%
PPO, fictitious play	**5%** 1% to 11%	**11%** 1% to 29%	**4%** 3% to 6%

Change in the win rate against the best response, in points, from "all data, one step" (rows in the order shown): mini-batches of 256: −11, −1, +0, +6; replay of 4 iterations: +0, −2, +37, −1. Single seeds of one setting differ by up to 32 points.

The exact solver's values as the critic: random board

A2C with a learned critic against A2C with the exact solver's values as a frozen critic.



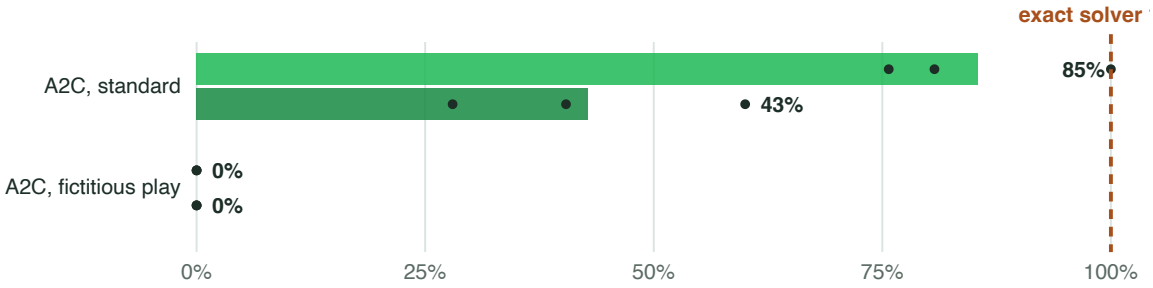
■ learned critic ■ exact critic

learner	learned critic	exact critic
A2C, standard	**14%** 11% to 19%	**10%** 0% to 26%
A2C, fictitious play	**12%** 8% to 14%	**4%** 1% to 11%

Change in the win rate against the best response, in points, from "learned critic" (rows in the order shown): exact critic: –5, –8. Single seeds of one setting differ by up to 25 points.

The exact solver's values as the critic: deterministic board

A2C with a learned critic against A2C with the exact solver's values as a frozen critic.



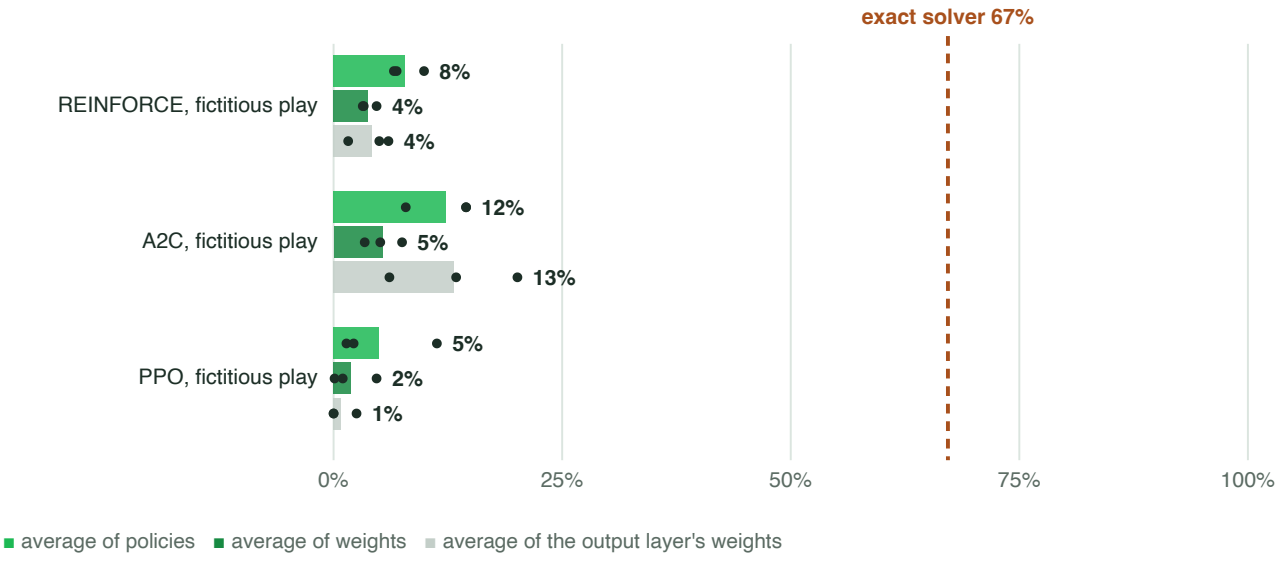
■ learned critic ■ exact critic

learner	learned critic	exact critic
A2C, standard	**85%** 76% to 100%	**43%** 28% to 60%
A2C, fictitious play	**0%** 0%	**0%** 0%

Change in the share tied with the exact Nash policy, in points, from "learned critic" (rows in the order shown): exact critic: –43, +0. Single seeds of one setting differ by up to 32 points.

Averaging weights instead of policies

Random board, fictitious play. Fictitious play averages policies; the other two build one network from the snapshots' weights (all layers, or only the output layer).



learner	average of policies	average of weights	average of the output layer's weights
REINFORCE, fictitious play	**8%** 7% to 10%	**4%** 3% to 5%	**4%** 2% to 6%
A2C, fictitious play	**12%** 8% to 14%	**5%** 3% to 8%	**13%** 6% to 20%
PPO, fictitious play	**5%** 1% to 11%	**2%** 0% to 5%	**1%** 0% to 2%

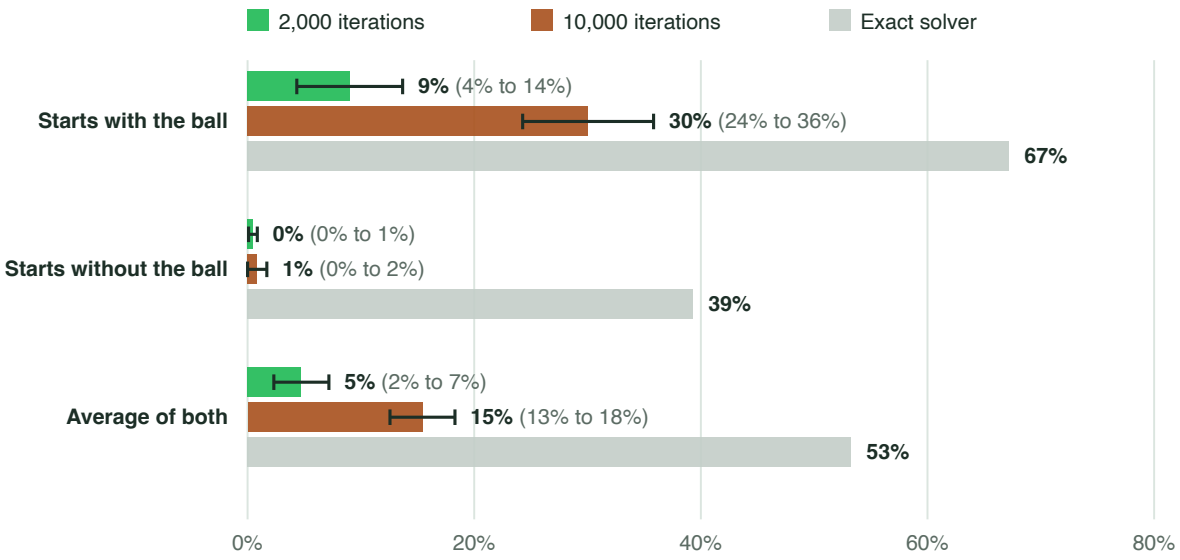
Change in the win rate against the best response, in points, from "average of policies" (rows in the order shown): average of weights: –4, –7, –3; average of the output layer's weights: –4, +1, –4. Single seeds of one setting differ by up to 14 points.

14. More seeds

With 3 seeds, A2C with fictitious play looked better at 10,000 iterations than at 2,000, and PPO with standard training looked better with replayed data. Each of these two settings and its usual-setting baseline was run on seeds 0 to 9 (random move-order board). Each bar is the mean of the 10 seeds and the thin line is its 95% interval (Student's t, kept within 0% and 100%). The exact solver wins 67% when it starts with the ball, 39% when it starts without the ball and 53% averaged over both starts.

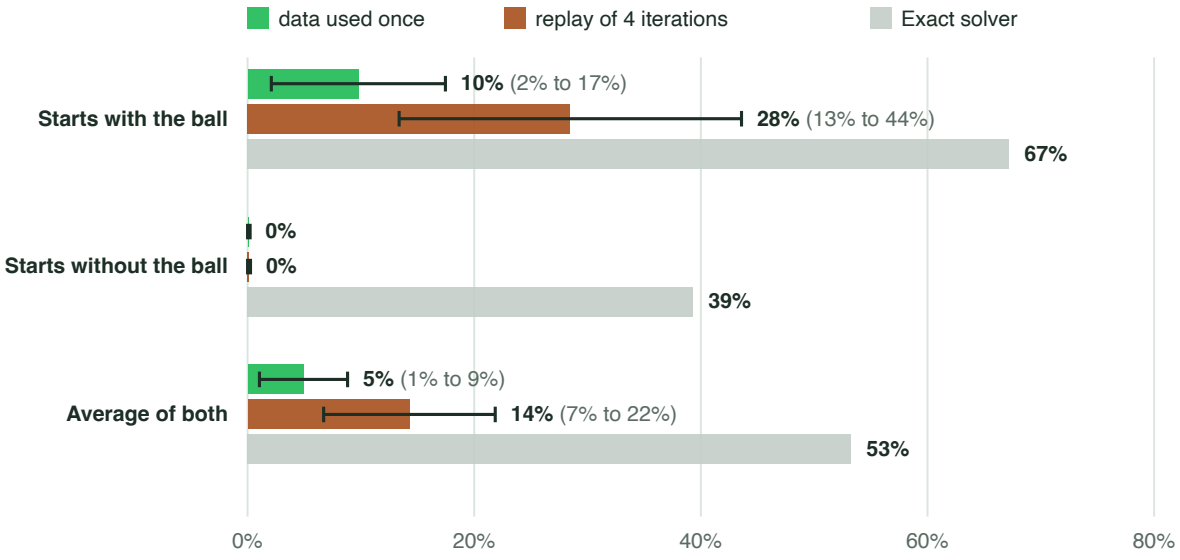
setting	starts with the ball (95% interval)	starts without the ball	average of both	single seeds, starts with the ball
A2C, fictitious play, 2,000 iterations	**9%** 4% to 14%	**0%** 0% to 1%	**5%** 2% to 7%	1% to 20%
A2C, fictitious play, 10,000 iterations	**30%** 24% to 36%	**1%** 0% to 2%	**15%** 13% to 18%	11% to 40%
PPO, standard, 2,000 iterations	**10%** 2% to 17%	**0%** 0%	**5%** 1% to 9%	0% to 29%
PPO, standard, replay of 4 iterations	**28%** 13% to 44%	**0%** 0%	**14%** 7% to 22%	1% to 62%

A2C, fictitious play: 2,000 iterations against 10,000 iterations



Starts with the ball 9% (4% to 14%) → 30% (24% to 36%), a change of +21 points (+14 to +28); averaged over both starts 5% → 15%, +11 points (+7 to +14); when it starts without the ball the change is +0 points (−1 to +1). The interval for the change when it starts with the ball excludes zero.

PPO, standard: data used once against replay of 4 iterations



Starts with the ball 10% (2% to 17%) → 28% (13% to 44%), a change of +19 points (+3 to +35); averaged over both starts 5% → 14%, +9 points (+1 to +17); when it starts without the ball the change is +0 points (+0 to +0). The interval for the change when it starts with the ball excludes zero.

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15. Are the two players mirror images?

The board is symmetric: swapping the two players and mirroring the pitch gives the same game. The **mirror gap** measures how far player 1's policy is from the mirror image of player 0's: for every position and several steps it takes the largest difference in move probabilities between player 0's policy and the mirrored policy of player 1, then averages. 0 means the pair is perfectly symmetric, 1 means opposite. The exact solver is symmetric (0). The learners are not: their two networks are trained separately and settle on different policies, with gaps of 0.15 to 0.81; fictitious play has the smaller gap in 7 of the 9 learner-and-board pairs.

policy	Deterministic board (A10)	Random move-order board	Continuing game
Exact solver	0.00	0.00	0.00
REINFORCE, standard	0.66	0.56	0.33
REINFORCE, fictitious play	0.44	0.29	0.39
A2C, standard	0.53	0.60	0.39
A2C, fictitious play	0.44	0.36	0.41
PPO, standard	0.81	0.70	0.67
PPO, fictitious play	0.48	0.59	0.44
A2C, exact critic, standard	–	–	0.38
A2C, exact critic, fictitious play	–	–	0.15

16. Other kinds of games

Best-response dynamics are known to converge in zero-sum games and in potential games, such as coordination games, but only to one of the equilibria and not necessarily from every start. Three two-by-two games show the difference. **Simultaneous**: both players switch to the best response to the other's last action. **Alternating**: one player responds, then the other responds to that. **Fictitious play**: each player best-responds to the average of the other's past play; the last column is the share of the first action played by the row and by the column player (2,000 rounds). Battle of the sexes (both prefer to be together; each prefers a different place): payoffs B,B = 2 and 1, S,S = 1 and 2, otherwise 0. Different choices: 1 each if the players choose differently, otherwise 0. Matching pennies: the row player wants to match, the column player wants to differ.

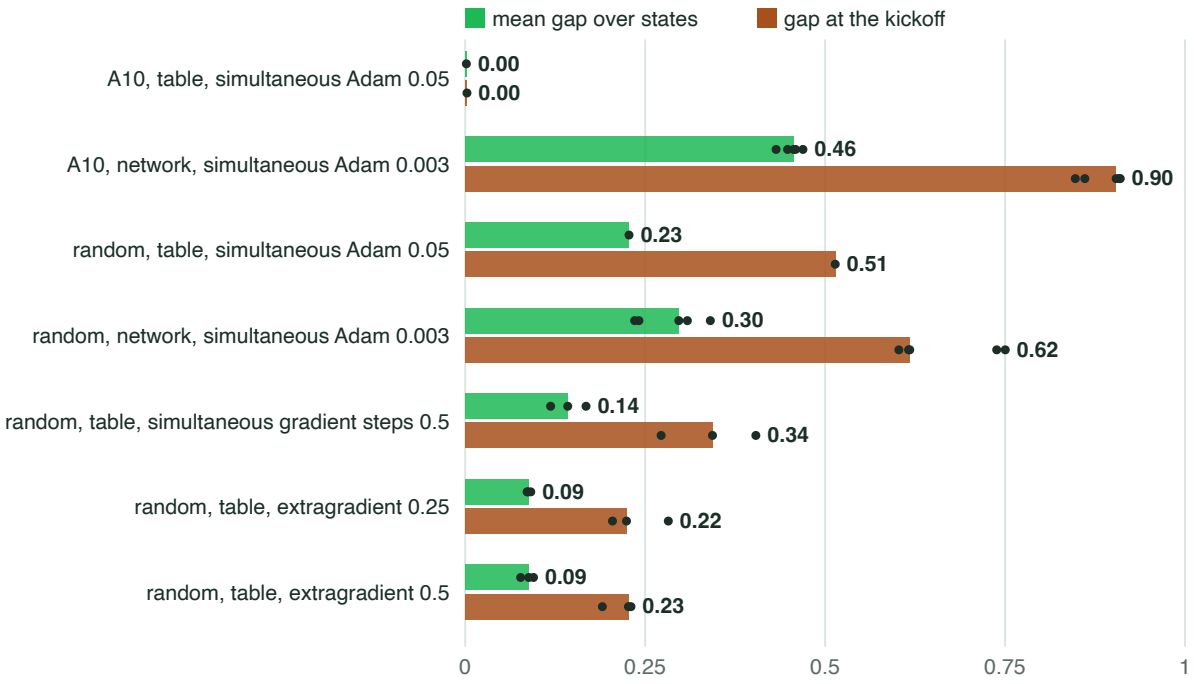
game	pure equilibria	mixed equilibrium (first action)	start	simultaneous best responses end in	alternating best responses end in	fictitious play
Battle of the sexes (coordination)	BB, SS	0.67 / 0.33	BB	BB	BB	1.00 / 1.00
			BS	keeps switching	SS	1.00 / 1.00
			SB	keeps switching	BB	1.00 / 1.00
			SS	SS	SS	0.00 / 0.00
Different choices (anti-coordination)	01, 10	0.50 / 0.50	00	keeps switching	10	0.50 / 0.50
			01	01	01	1.00 / 0.00
			10	10	10	0.00 / 1.00
			11	keeps switching	01	0.50 / 0.50
Matching pennies (zero-sum)	none	0.50 / 0.50	HH	keeps switching	keeps switching	0.50 / 0.49
			HT	keeps switching	keeps switching	0.50 / 0.51
			TH	keeps switching	keeps switching	0.50 / 0.51
			TT	keeps switching	keeps switching	0.50 / 0.51

In the coordination game the dynamics end in one of the two pure equilibria, which one depending on the start; started with mismatched choices the simultaneous version keeps switching. Where the only equilibrium is mixed (matching pennies) best-response dynamics never settle, but fictitious play's averages go to one half each, as in rock-paper-scissors. In the game of different choices, started with the same choice, simultaneous best responses keep switching and fictitious play's averages go to one half each while the players always match; started with different choices it stays there.

17. Without sampling noise: exact gradient play

To separate the learning rule from the noise of sampled gradients, both players follow the **exact** gradient of their own value, computed from the transition table on the stationary discounted game (discount 0.9, no step count) and summed over every state, as if a game could start anywhere. The **duality gap** at a state is the first player's best-response value against the second's policy minus the second's best-response value against the first's; it is 0 only at a Markov-perfect (stationary Nash) equilibrium. A **table** has one row of move logits per state; a **network** is one 64×64 network per player, shared across states.

Bars are the median over runs (1 table run, 5 network seeds, 3 seeds for the 6,000-step runs) and dots single runs (`scripts/pg_exact_play.py`).



- **Deterministic board (pure equilibrium everywhere): the table reaches the equilibrium and the networks do not.** The table ends with a mean gap of 0.002 and a largest gap of 0.008; every network stalls with a mean gap of 0.43–0.47 and a kickoff gap of 0.85–0.91, the same scale as the sampled learners' exploitability. On this board the network, not the sampling noise, is what stops policy gradient.
- **Random board (the equilibrium mixes at 94 states): no exact gradient method tried here reaches it, even with a table.** The table with Adam stays at a mean gap of 0.23; simultaneous gradient steps leave 0.12–0.17 after 6,000 steps; extragradient, a standard fix for the cycling of simultaneous gradient play near mixed equilibria, roughly halves that to 0.077–0.095 and then stops improving. The networks end at 0.24–0.34.
- So there are two separate obstacles: sharing a network across states stops learning even where the equilibrium is pure, and where it has to mix, simultaneous gradient play does not settle even with a table. Why extragradient stops improving is not tested. The game here is the stationary discounted one, not the finite-horizon games of the sections above.

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18. Related work

Has fictitious play been combined with policy gradient before? Yes, in several forms, and the convergence theory is much older than the neural networks. I checked the titles, authors and venues against Google Scholar (its top result for fictitious play, policy gradient and zero-sum Markov games was Liu, Yang and Wang) and compared each one-line summary with the abstract on the paper's own proceedings, journal or arXiv page, or on a bibliographic mirror where the publisher's page would not load (noted in the entry). JSTOR and the publisher pages for the 1951 papers would not load, so Brown and Robinson are cited from the linked secondary source and were not read. I have not checked the proofs. I did not find a result that proves convergence for deep policy gradient with fictitious play in zero-sum Markov games; the closest results are tabular, regularized or for special game classes.

Older work

- Iterative solution of games by fictitious play (Brown, 1951; Activity Analysis of Production and Allocation, ed. T. C. Koopmans, Wiley, pp. 374-376). Introduced fictitious play as a method for solving two-person zero-sum games (first proposed in 1949); generally regarded as the origin of the method. Not read directly: the publisher and JSTOR pages did not load, so the summary and pages (374-376) come from the linked paper and a university lecture reference list.
- An iterative method of solving a game (Robinson, 1951; Annals of Mathematics 54(2), pp. 296-301). Shows that the iterative procedure suggested by Brown, in which each player in turn plays the best pure strategy against the opponent's accumulated mixed strategy, converges for two-person zero-sum (matrix) games; Shapley (1964) later showed that convergence fails in general games. Not read directly (JSTOR did not load): the summary comes from a Semantic Scholar record and the linked paper.

- Markov games as a framework for multi-agent reinforcement learning (Littman, 1994; ICML, pp. 157-163). Extends reinforcement learning from one adaptive agent to several; studies two agents with directly opposed goals sharing an environment and gives a Q-learning-style algorithm (minimax-Q), shown on a simple game whose best strategy is probabilistic.
- Fictitious play property for games with identical interests (Monderer and Shapley, 1996; Journal of Economic Theory 68(1), pp. 258-265). Defines n-person games with identical interests (best-response equivalent in mixed strategies to a game with identical payoff functions) and proves that every such game has the fictitious-play property. Abstract read on a bibliographic mirror; the publisher's page was blocked.
- Potential games (Monderer and Shapley, 1996; Games and Economic Behavior 14(1)). Defines several notions of potential function for games in strategic form and characterizes the games that have one (in the standard sense that a single function captures every player's incentive to change strategy). Abstract read on a copy of the paper's first page.
- Multiagent learning using a variable learning rate (Bowling and Veloso, 2002; Artificial Intelligence 136(2)). Multiagent learning in stochastic games: a reinforcement-learning technique with a variable learning rate (WoLF, win or learn fast), with a convergence proof in self-play for a restricted class of iterated matrix games and experiments on more general stochastic games.
- On the global convergence of stochastic fictitious play (Hofbauer and Sandholm, 2002; Econometrica 70(6)). Stochastic (perturbed) fictitious play converges globally in four classes of games: those with an interior evolutionarily stable state, zero-sum games, potential games and supermodular games.
- Nash Q-learning for general-sum stochastic games (Hu and Wellman, 2003; Journal of Machine Learning Research 4). Q-learning for general-sum Markov games in which each agent assumes Nash play on the current Q-values; proved to converge under restrictions on the intermediate stage games, which grid-game experiments suggest are not always necessary.
- Generalised weakened fictitious play (Leslie and Collins, 2006; Games and Economic Behavior 56(2)). A broad class of adaptive learning processes extending weakened fictitious play, shown to converge in games with the fictitious-play property; includes an actor-critic process in which a player sees only the reward of the action it chose. Abstract read on the authors' Bristol page.
- Brown's original fictitious play (Berger, 2007; Journal of Economic Theory 135(1)). Brown's 1951 process updates beliefs in a different order from the version used today; with that order, a short convergence proof for nondegenerate ordinal potential games.

Recent work

- Fictitious Self-Play in Extensive-Form Games (Heinrich, Lanctot and Silver, 2015; ICML, PMLR 37). Fictitious play in behavioural strategies of extensive-form games, and a sample-based version (fictitious self-play); approximate Nash equilibria in poker games.
- Deep Reinforcement Learning from Self-Play in Imperfect-Information Games (Heinrich and Silver, 2016; arXiv). Neural fictitious self-play (NFSP) combines fictitious self-play with deep reinforcement learning to approximate Nash equilibria end to end; on Leduc poker it converged near an equilibrium where standard reinforcement learning methods failed to converge, and on Limit Texas Hold'em it came close to leading algorithms that rely on expert knowledge.
- Independent Policy Gradient Methods for Competitive Reinforcement Learning (Daskalakis, Foster and Golowich, 2020; NeurIPS 33). Both players running policy gradient simultaneously reach a min-max equilibrium of a two-player zero-sum stochastic game, provided the learning rates follow a two-timescale rule (which the authors show is necessary).
- Competitive Policy Optimization (Prajapat, Azizzadenesheli, Liniger, Yue and Anandkumar, 2021; UAI 37, PMLR 161, pp. 64-74). Tests game-aware competitive policy gradient (CoPG) and trust-region CoPO against standard gradient-descent-ascent in Markov soccer; CoPG wins more than 74% of reported games against GDA. This is an empirical result for the paper's soccer setup, not a proof of Nash convergence or a training-time guarantee for this project.
- Policy Optimization in Zero-Sum Markov Games: Fictitious Self-Play Provably Attains Nash Equilibria (Liu, Yang and Wang, 2020; OpenReview). A smooth fictitious self-play in which each player takes an entropy-regularized proximal policy optimization (PPO) step against the other; the joint policies converge to a neighbourhood of a Nash equilibrium at a sublinear rate, the first finite-time guarantee for fictitious self-play in zero-sum Markov games (for the regularized method, not the exact equilibrium). Authors and year from the Google Scholar listing and a citation in another paper; abstract read on the Semantic Scholar listing, as the paper's own page did not load.
- Provably Efficient Fictitious Play Policy Optimization for Zero-Sum Markov Games with Structured Transitions (Qiu, Wei, Ye, Wang and Yang, 2021; ICML, PMLR 139). Fictitious-play policy optimization with UCB-style optimism for two-player zero-sum Markov games with structured, unknown transitions (factored independent or single-controller), with tight regret bounds of order the square root of the number of rounds.
- Policy Optimization for Markov Games: Unified Framework and Faster Convergence (Zhang, Liu, Wang, Xiong, Li and Bai, 2022; NeurIPS 35). Each iteration updates the policy at every state with a matrix-game algorithm and then the values; the state-wise average policy converges to an approximate Nash equilibrium of two-player zero-sum Markov games when the matrix-game learners have low weighted regret (rates of roughly T^-5/6, and T^-1 with a modified value update).
- Fictitious play in zero-sum stochastic games (Sayin, Parise and Ozdaglar, 2022; SIAM Journal on Control and Optimization 60(4), pp. 2095-2114). Combines classical fictitious play with Q-learning in two-player zero-sum stochastic games; the strategy beliefs converge to a stationary mixed Nash equilibrium, in both the model-based and model-free cases.
- Fictitious Play in Markov Games with Single Controller (Sayin, Zhang and Ozdaglar, 2022; ACM EC). A new variant of fictitious play for Markov games converges to a Nash equilibrium in n-player identical-interest games where a single player controls the transitions; together with earlier results it covers both zero-sum and single-controller identical-interest Markov games.
- Fast Policy Extragradient Methods for Competitive Games with Entropy Regularization (Cen, Wei and Chi, 2024; Journal of Machine Learning Research 25(4)). Extragradient methods reach the entropy-regularized equilibrium at a linear rate (also in zero-sum Markov games), and tuning the regularization weight approaches an approximate Nash equilibrium of the unregularized game at a sublinear rate.

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19. Reproducing the numbers

Every number in this report is computed by `scripts/pg_site_data.py` from the result files in `experiments/`; the same data feeds the website's Policy gradient tab. The runner is `scripts/pg_finite.py` (2,000 iterations of 64

episodes and 1,000 evaluation games by default). The saved comparisons used one REINFORCE/A2C update pass per collected batch; the runner now defaults to five and writes those runs to separate files. Use `--updates-per-batch 1` to reproduce the saved data. PPO retains four epochs. Each output row records the effective update count. `make pg-report` and `make pg-pdf` rebuild the short report and `make pg-detailed-pdf` this one.

file	command	what	time
pg_finite_a10.csv	pg_finite.py --seeds 3 --updates-per-batch 1	historical discounted 100-step results: REINFORCE / A2C / PPO by self-play and fictitious play vs. exact finite-horizon solution, **3 seeds**; the runner now defaults to five REINFORCE/A2C updates and writes _updates5.csv separately	~40 min
pg_finite_rate_a10_*.csv	pg_finite.py --scoring rate --algos <algo> --modes <mode> --tag _<algo>_<mode>	the continuing game (play restarts after a goal, fixed 100 steps; a win is more goals), including A2C with the exact solver's values as a frozen critic (a2c_exact); **3 seeds**	~30 min each
pg_variant_{random,a10,rate_a10}_{shared,trim10}.csv	pg_finite.py --board <random|a10> [--scoring rate] --algos a2c ppo --iterations 2000 --shared --tag _shared (and --trim 10 --tag _trim10)	A2C and PPO with one shared network (two output heads), and with the last 10 steps of each episode left out of the loss, on all three boards; **3 seeds**	~1 h each
pg_variant_{random,a10}_argmax_{reinforce,a2c,ppo}.csv	pg_finite.py --board <random|a10> --algos <algo> --modes fictitious_argmax --iterations 2000 --tag _argmax_<algo>	fictitious play whose snapshots are pure (argmax) policies, for the opponent mixture and the reported policy; **3 seeds**	~1–2 h each
pg_variant_rate_a10_argmax_{reinforce,a2c,ppo}.csv	pg_finite.py --board a10 --scoring rate --algos <algo> --modes fictitious_argmax --prefix pg_variant --tag _argmax_<algo>	the same argmax fictitious play on the continuing game; **3 seeds**	~1–3 h each
pg_entropy_{random,a10,rate_a10}_{selfplay,fictitious,fictitious_argmax}_{reinforce,a2c,ppo}.csv	pg_finite.py --board <random|a10> [--scoring rate] --algos <algo> --modes <mode> --entropy 0.2 --prefix pg_entropy --tag _<mode>_<algo>	every learner and training scheme on every board with the entropy bonus raised from 0.01 to 0.2 (an ablation); **3 seeds**	~5 min to ~1 h each
pg_study_{random,a10,rate_a10}_<setting>_<mode>_<algo>.csv	pg_finite.py --seeds 3 --resume --prefix pg_study --board <board> [--scoring rate] --algos <algo> --modes <mode> <option> --tag _<setting>_<mode>_<algo> with <option> one of --trim 5/20/40, --snap-every 1/25/100, --iterations 10000, --minibatch 256, --replay 4, --updates-per-batch 1, or --algos a2c_exact, and --weights for the weight-averaged networks	the training-choice comparisons (how many last steps to leave out, snapshot interval, five times the iterations, mini-batches, replayed data, repeated optimization updates, exact critic on every board, averaging weights); **3 seeds** each	~10 min to ~4 h each
pg_more_random_<setting>_s<seed>.csv	pg_finite.py --seeds 1 --seed-start <seed> --resume --prefix pg_more --board random --algos <a2c|ppo> --modes <fictitious|selfplay> [--iterations 10000 | --replay 4] --tag _<setting>_s<seed> for seeds 3–9, with seeds 0–2 from the earlier files	the two settings that looked best on 3 seeds (A2C fictitious play at 10,000 iterations, PPO standard training with replay of 4 iterations) and their baselines, on **10 seeds** each	~5 min to ~2 h each
pg_curves_{random,a10}_<mode>_<algo>.csv	pg_curves.py --board <board> --algos <algo> --modes <mode> --tag _<mode>_<algo>	win rate against the best response, and the kickoff policy of the current and reported network, every 100 iterations of training; **3 seeds**	~2–3 h each
pg_study_random_ep{8,256}_<mode>_<algo>.csv	pg_finite.py --seeds 3 --resume --prefix pg_study --board random --algos <a2c|ppo> --modes <mode> --episodes <8|256> --tag _ep<N>_<mode>_<algo>	A2C and PPO, standard and fictitious play, with 8 or 256 episodes per iteration instead of 64; **3 seeds**	~10 min to ~2 h each
pg_exact_play.csv	pg_exact_play.py (or --only a10, --only random, --only random-eg, --shard i/n)	exact (noise-free) gradient play on the stationary discounted game, scored by the exact duality gap every 100 or 250 steps: a table of logits (1 run) and a 64×64 network per player (**5 seeds**) with simultaneous Adam on both boards, and on the random board a table with simultaneous gradient steps or extragradient from a small random start, **3 seeds**, 6,000 steps	~20 min to ~2 h each
rps_nn_phases.json	rps_nn_phases.py	rock-paper-scissors networks with a snapshot every 5, 50 or 200 iterations (softmax and argmax averages); **3 seeds**, 8,000 iterations	~1 h
pg_exact_kickoff.json	pg_exact_kickoff.py	the exact solver's kickoff move probabilities on both boards, for the curve plots	~2 min
rps_nn.json	rps_nn.py	rock-paper-scissors with neural-network players: standard, fictitious play, argmax fictitious play, at the soccer settings and with a larger entropy bonus; **3 seeds**, 3,000 iterations	~15 min

pg_random_kickoff_value.json	pg_kickoff_value.py	value of the random board at the kickoff for the exact 100-step solution (the ball-holding seat's edge)	~1 min
pg_mixed_states.json	pg_mixed_states.py --algo <algo> --mode <mode> for each of the six learners, then --merge	the learners at the random board's 94 mixed states: win rates from games started there, distance from the exact mix, equilibrium regret, three states in full; **3 seeds**	~25 min each
pg_finite_random_{reinforce,a2c,ppo}.csv	pg_finite.py --board random --algos <algo> --tag _<algo>	the same six learners on the random move-order board (94 mixed stage games), discounted 100-step objective, **3 seeds**	~30 min each
pg_finite_a10_long_*.csv	pg_finite.py --iterations 8000 --modes selfplay --seeds 2	the same self-play learners at 4x the training, **2 seeds**	~15–35 min each
pg_fp_br_*.csv	pg_fp_br.py --algo a2c	fictitious play with best-response phases (100 or 300 policy-gradient iterations per best response), exploitability every 5 rounds	~8 min
pg_policy_outputs.json	pg_policy_outputs.py --algo a2c --mode selfplay, then --merge	action probabilities (U, D, L, R) of the six learners at four fixed positions, shown on the site and in the PDF	~3 min each